

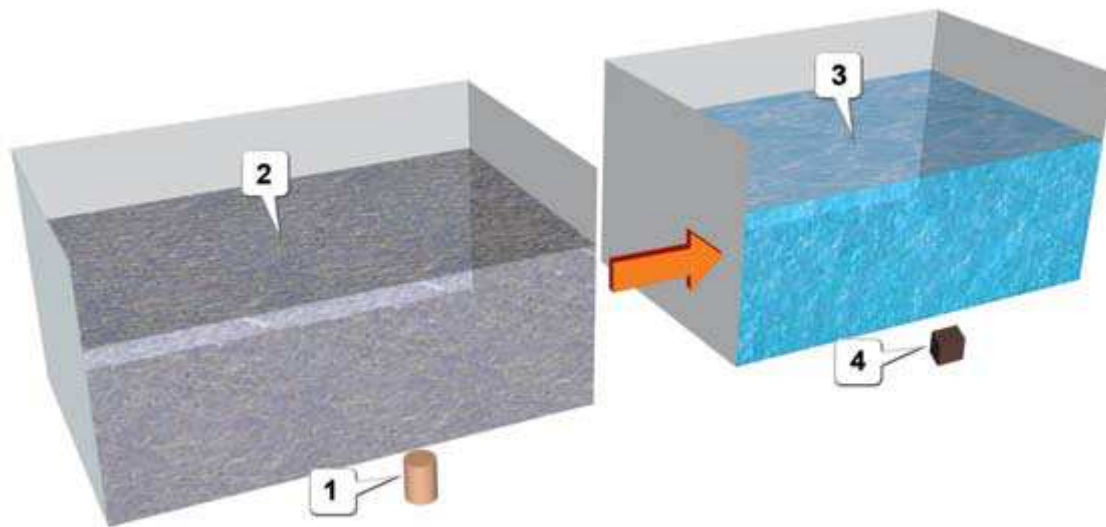
SEWAGE TREATMENT GUIDE/MANUAL

BY

CLEAR –ION EXPERT PVT. LTD.

The Operating Principle of STPs

First of all, let us understand the underlying concept of a biological sewage treatment plant.



Conceptually, the process is extremely simple: A small amount of microorganisms 1 converts a large mass of polluted water 2 into clean water 3. This process also produces a co-product: A vastly reduced, compact solid biomass 4 (the excess microorganisms produced by growth and multiplication of the original population of microorganisms).

However, translating this simple principle into a properly designed and engineered STP is a real challenge: It requires sound knowledge of the biology of the microorganisms, chemical and mechanical engineering principles, and an equally large dose of common sense.

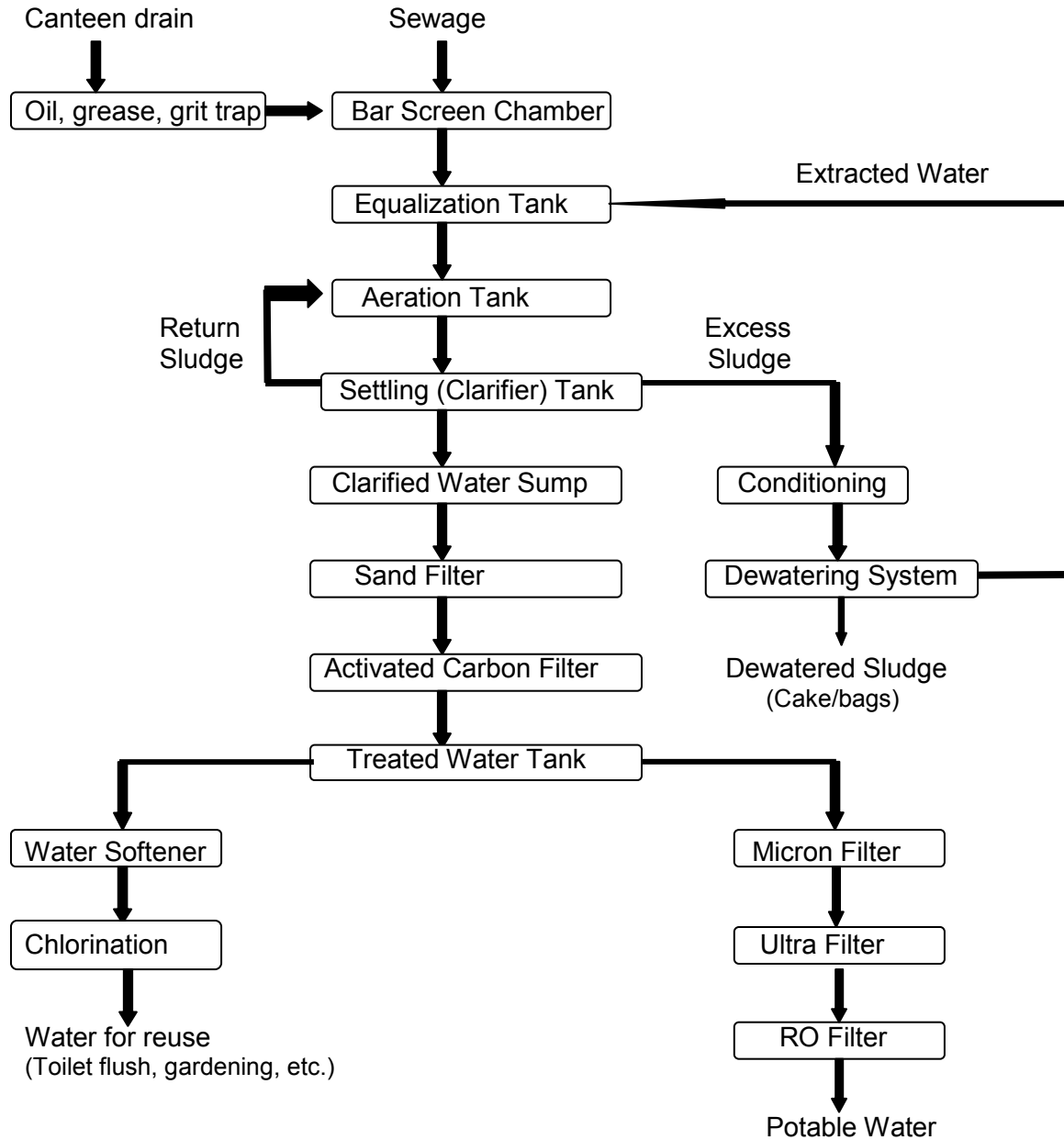
We need an STP that-

- Achieves the desired results on a consistent and sustained basis.
- Is robust and reliable, and lasts for at least 10-15 years without major repairs.
- Needs minimum amounts of money, energy and chemicals to achieve the desired treated water quality.
- Is easy to operate and maintain.

This manual provides tips on how to build and operate such an STP.

Typical Process in an STP

The flow chart of a typical STP is shown below (optional units are shown in yellow).



Benefits of a well-run STP

The primary benefits of a well-run STP are-

- ❖ Assured availability of water for various secondary uses
- ❖ Enormous savings in fresh water costs¹
- ❖ Lesser Environmental Degradation
- ❖ Improved public Health

The following table illustrates the quality of water obtainable from a well-designed, engineered and operated STP at very affordable treatment costs².

Parameter	In raw sewage	After treatment	What treatment to you
pH	6.5-7.5	6.5-7.5	The acidity/alkalinity balance is not affected/alterd.
BOD	200- 250 mg/L	< 10 mg/L	Normally, the biodegradable material in the sewage consumes oxygen when it degrades. If this sewage is released in lakes/rivers, it would draw naturally dissolved oxygen from water, depleting the oxygen in the lake/river. This causes death of fish and plants. But the STP provides enough oxygen to digest the biodegradable material in sewage. The treated sewage does not need oxygen any longer. Thus it does not affect the aquatic life in lakes and rivers.
Turbidity	Not specified	< 10 NTU ₂	The outgoing treated sewage has low turbidity (suspended particles that cloud the water). In other words, we get "clear" water. This prevents the pipelines from getting clogged by settled sediments. If cloudy water is allowed to reach the lakes and rivers, it blocks the sunlight from reaching the bottom of the water body. This stops the photosynthesis process of the aquatic plants, killing them. That in turns stops generation of oxygen as a byproduct of the photosynthesis process. Depletion of dissolved oxygen in water kills all fish. Thus low turbidity in discharge water ultimately sustains aquatic life in lakes and rivers.
E. Coli	Not specified	NIL	The STP removes the harmful bacteria completely.

Understanding the STP Stages

Bar Screen Chamber

1.1 Function

The function of the bar screen is to prevent entry of solid particles/ articles above a certain size; such as plastic cups, paper dishes, polythene bags, condoms and sanitary napkins into the STP. (If these items are allowed to enter the STP, they clog and damage the STP pumps, and cause stoppage of the plant.)

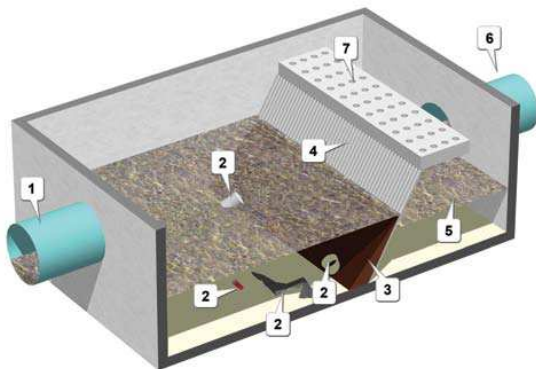
The screening is achieved by placing a screen made out of vertical bars, placed across the sewage flow.

- The gaps between the bars may vary between 10 and 25 mm.
- Larger STPs may have two screens: A coarse bar screen with larger gaps between bars, followed by a fine bar screen with smaller gaps between bars.
- In smaller STPs, a single fine bar screen may be adequate.

If this unit is left unattended for long periods of time, it will generate a significant amount of odor: it will also result in backing of sewage in the Incoming pipelines and chambers.

1.2 How It Works

A typical Bar Screen Chamber (also called a "Bar Screen Channel") is shown here (cutaway view).



S. no	Remark
1.	Inlet pipe for the STP.
2.	Debris (plastic bags, paper cups, condoms, sanitary napkins, paper dishes, etc.) gets trapped here.
3.	Muck (sediment in sewage) accumulates and blocks the grill (if not cleaned regularly)
4.	Grill. Must be cleaned regularly to avoid a build-up of debris (2) and muck (3).
5.	Screened sewage. If the screen (4) is maintained well, this would be free of any large articles.
6.	Outlet pipe (goes to the Equalization Tank)
7.	Platform with weep holes. The STP operator stands here to rake the debris (2). He also uses the platform as a drip-tray for the collected debris

1.3 Design Criteria

The design criteria applies more to the sizing and dimensions of the Screen chamber rather than the screen itself.

- | | |
|----|---|
| 1. | The screen chamber must have sufficient cross-sectional opening area to allow passage of sewage at peak flow rate (2.5 to 3 times the average hourly flow rate) at a velocity of 0.8 to 1.0 m/s, (The cross-sectional area occupied by the bars of the screen itself is not to be counted in this calculation.) |
| 2. | The screen must extend from the floor of the chamber to a minimum of 0.3 m above the maximum design level of sewage in the chamber under peak flow conditions. |

1.4 Construction And Engineering

Bar screen racks are typically fabricated out of 25mm x 6 mm bars either of epoxy-coated mild steel or stainless steel. A specified opening gap is kept between the bars. The screen frame is fixed in the bar screen chamber at an angle of 60° to the horizontal, leaning away from the incoming side. Care is to be taken to see that there are no gaps left between the screen frame and the floor and the sides of the chamber. The upper end of the screen must rest against an operating platform, on which the STP operator stands to rake the debris collected at the grill. The platform itself must be provide with weep holes. So that the operator can leave the collected debris on the platform for the screened debris to drip back into the chamber. This not only reduces the weight and volume of trash to be finally disposed off, but also reduces the nuisance of odor coming from the putrefying matter.

1.5 Operation And Maintenance Considerations

- Check and clean the bar screen at frequent intervals
- Do not allow solids to overflow/ escape from the screen
- Ensure no large gaps are formed due to corrosion of the screen
- Replace corroded/ unserviceable bar screen immediately

1.6 Troubleshooting

Problem	Cause
Large articles pass through, and choke the pumps	Poor design / poor operation / screen damaged
Upstream water level is much higher than downstream level	Poor operation (inadequate cleaning)
Excessive collection of trash on screen	Poor operation
Excessive odor	Poor operation / trash disposal practices

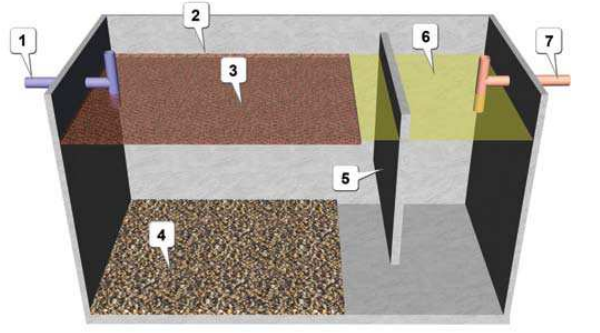
Oil And Grease/Grit Trap

2.1 Function

The grease and grit trap is placed at the discharge point of the canteen/ kitchen area itself to arrest solid and fatty matter at source. The wastewater output from this unit is taken to the equalization tank. The solids and fats that are separated in this unit are disposed off along with other biodegradable waste, and can be used as feed for piggeries. Separating solids (rice, vegetables, pulses) and grease from the wastewater at source ensures that the contact time between solids and wastewater is kept to a minimum, so that the wastewater does not absorb additional organic pollutant loads (starch, carbohydrates, proteins) due to leaching of these substances from the solids. (Rather than building a larger STP to digest this extra organic matter, it is far more economical to prevent the organic matter from entering the STP.) An Oil and grease/grit trap is generally not an essential unit in a typical residential complex. It is however a mandatory unit in commercial and Industrial units with a canteen on campus.

2.2 How It Works

A typical Oil and grease/grit trap is shown below (the front side is removed to show internal structure).

	S. no	Remark
	1.	The incoming liquid is released below surface through a T-joint so that the falling water does not disturb (break up) the floating film of fat and scum (3).
	2.	The tank is always filled till this level .
	3.	The fat and scum rise to the top and float on the liquid. This needs to be removed periodically, otherwise it will leach into the wastewater.
	4.	The heavier grit and solids sink to the bottom of the tank (most of it lies below

		the inlet pipe, but some of the grit may be moved toward the outlet side due to the strong flow of the wastewater). This mass also needs to be removed from the tank periodically.
	5.	The baffle plate prevents the floating fat and scum (3) from drifting towards the outlet (7).
	6.	Wastewater reaching the outlet side is free of fat, scum, grit and solids
	7.	The outlet is through a T-joint pipe, similar to the inlet (1). The upper part is capped off (opened only for maintenance).

2.3 Design Criteria

Typical design criteria used for the grease trap include:

1.	Shallow trap (to allow quick rise of oils and fats to the surface)
2.	The length of trap should be approximately 2 times its depth
3.	Residence time in the trap is optimally 5-20 minutes at peak flow. (Increasing the time does not result in appreciable improvement)
4.	Surface area of the trap in m ² should be approximately 1.5 to 2 times the depth of trap in meters.

2.4 Construction And Engineering

The tank should have waterproof plastering inside and out. The end of the incoming pipe is kept below the water level, so that the incoming water does not disturb (and break up) the upper floating layer of grease. The trapped material (both floating film of grease/ fat and the grit settled at bottom) must be collected frequently; otherwise the trap will fail to serve its fundamental purpose. Therefore the trap must be engineered to facilitate frequent removal of these two layers. For example, the covers must be made of lightweight materials for easy lifting. Large traps may be provided with vent pipes to release gases.

2.5 Operation And Maintenance Considerations

- Check and clean trap at frequent intervals
- Remove both settled solids (at bottom) and the floating grease
- Do not allow solids to get washed out of the trap
- Redesign the trap if solids and grease escape on a regular basis, despite good cleaning practices

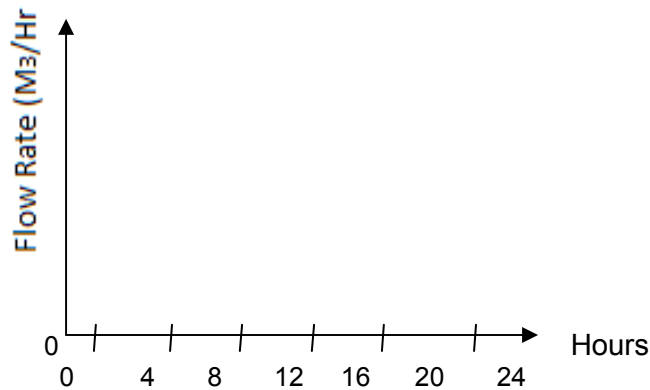
2.6 Troubleshooting

Problem	Cause
Oil and grease pass through the trap	Poor design/ poor operation
An excessive amount of solids passes through the trap	Poor design/ poor operation
Excessive odor	Poor operation/ waste disposal practices

Equalization Tank

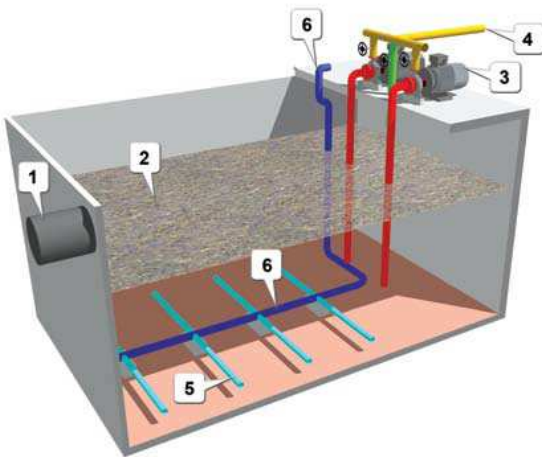
3.1 Function

The sewage from the bar screen chamber and oil, grease and grit trap comes to the Equalization tank. The equalization tank is the first collection tank in an STP. Its main function is to act as buffer: To collect the incoming raw sewage that comes at widely fluctuating rates, and pass it on to the rest of the STP at a steady (average) flow rate. During the peak hours, sewage comes at a high rate. The equalization tank stores this sewage, and lets it out during the non-peak time when there is no/little incoming sewage. Thanks to the constant outflow rate, it is easier to design the rest of the units of the STP.



3.2 How It Works

A typical equalization tank is shown here.



Notes:

1. The figure uses color-coding only to distinguish the parts from each other: In real life the color-coding is not followed.
2. An air-compressor is required, but not shown because in most cases a single blower provides the compressed air needed at multiple places in the STP.
3. The figure shows only the surface of sewage (2), so that other items submerged in the sewage can be shown.

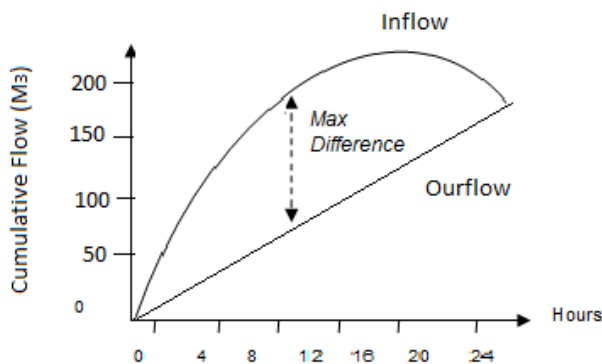
S. L	Remark
1.	The inlet pipe carries filtered sewage from the Bar Screen Chamber.
2.	The sewage is collected in the tank. The level fluctuates throughout the day, because while the incoming rate fluctuates widely, the out - going rate is constant. (The level shown in the figure is almost full. If there is a peak inflow now, the tank will overflow.
3.	The raw sewage lift pumps move the sewage to the aeration tank. (These pumps are explained in the next chapter.)
4.	The delivery pipe takes the sewage to the aeration tank.
5.	The coarse bubble diffusers are short length of tubes that have holes at regular spacing. They release large bubbles in the tank to lightly aerate the sewage, and also to agitate the mix continuously. The figure shows an array of eight diffusers, strapped to cement blocks so that the assembly remains firmly anchored in one place. Diffusers can also be used in separate pairs or even individually.
6.	Compressed air comes through this air supply pipeline. This may be a rigid pipe or a flexible hose. The figure shows a single array of 8 diffusers. However, it is more convenient to use separate pairs of diffusers with their own air pipe(flexible hose).

3.3 Design Criteria

Since the diurnal variation in the quality of the sewage is not significant, the equalization tank is used only for buffering the daily fluctuations in the sewage flow quantity. The equalization tank must be of sufficient capacity to hold the peak time inflow volumes. Peak times and volumes are site-specific and variable:

- In the case of residential complexes, there is a distinct morning major peak (when all residents are using their kitchens, bathrooms and toilets), followed by a minor peak in the late evening hours. In a typical residential complex, an equalization tank with a capacity to hold 4-6 hours of average hourly flow should be adequate (based on the diversity of the population in the complex).
- In addition, the sewage generation may be heavier during the weekends. In such cases, the sewage volume generated on a weekend should be taken as reference.
- In the case of a commercial or software complex, peak flows commonly occur during the lunch hour.
- In the case of manufacturing units, the shift timings is a major factor. Peaks occur at breakfast, lunch and dinner timings of the canteen.

A fairly scientific method of calculating the required capacity of the Equalization tank is by plotting a graph of the projected inflow and outflow over a 24-hour period, as shown below. The equalization tank should be large enough to hold the maximum difference between the inflow and the outflow. In our example, the maximum difference is $150 - 60 = 90 \text{ m}^3$. Therefore, the equalization tank must be larger than 90 m^3 (otherwise it will overflow).



3.4 Construction And Engineering

The incoming sewer line is usually gravity-fed, and is likely to be at considerable depth below the ground level. Therefore it is prudent not to make the tanks of STP too deep, otherwise it requires very deep excavations and expensive construction. It also makes the maintenance and cleaning processes very hazardous. In it necessary to force compressed air in the sewage held in the tank. This is mandatory for two reasons:

- It keeps the raw sewage aerated, thereby avoiding simplicity and suppressing odor generation.
- It keeps solids in suspension and prevents settling of solids in the tank, thereby reducing frequency of manual cleaning of the tank

The tank may be of any shape, provided it permits placement of air diffusers for full floor coverage and uniform mixing over the entire floor area. The diffusers should be retrievable: Individual diffusers (or sets of diffusers) may be lifted out and cleaned for routine maintenance. This will reduce frequency of shut down of the Equalization tank for manual cleaning purposes. If membrane diffusers are used, they will fail frequently, due to the repeated cycles of expansion and contraction caused by fluctuating water levels in the equalization tank. Therefore, only **coarse bubble diffusers** must be used in the equalization tank.

As a rule of thumb, the higher of the following two figures is taken as the air volume required per hour:

- 1.2-1.5 times the volume of the Equalization tank, or
- $2.5 - 3.0 \text{ m}^3/\text{m}^2$ of floor area.

The number and placement of diffusers must be adequate to dispense the calculated amount of air in the tank. The capacity of the air blower must be adequate to deliver the required quantity of air to the equalization tank as well as all other aerated tanks it serves. This tank is most prone to odor generation, since it contains raw (untreated) sewage. It may also build up gas, which can be explosive. Therefore it must have good ventilation

3.5 Operation And Maintenance Considerations

- Keep air mixing on at all times
- Ensure that the air flow/ mixing is uniform over the entire floor of the tank. Adjust the placement of diffusers and the air-flow rate as needed.
- Keep the equalization tank nearly empty before the expected peak load hours (otherwise it will

overflow)

- Check and clean clogged diffusers at regular intervals
- Manually evacuate settled muck/ sediments at least once in a year

3.6 Troubleshooting

Problem	Cause
Insufficient mixing/aeration	Poor design, engineering
Excessive odor	Poor design, engineering
Insufficient capacity to handle peak flows	Poor design
Usable capacity reduced due to solids accumulation	Poor maintenance

Raw Sewage Lift Pumps

4.1 Function

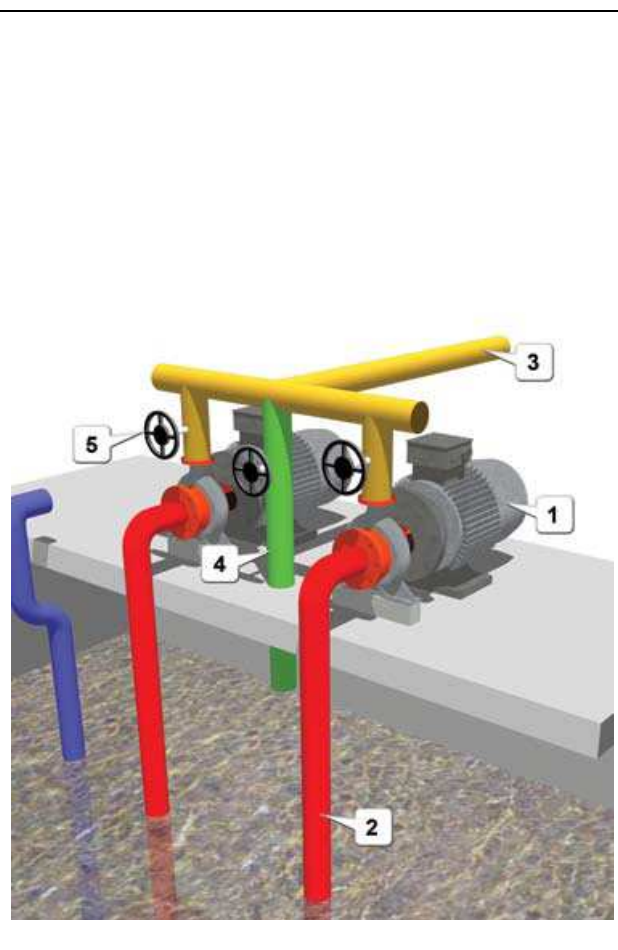
If we use gravity to move the sewage through the units of STP, the units would have to be placed progressively deeper below the ground level. To avoid deep excavations, a pumping stage is introduced to lift sewage to the next unit in the STP, which is the **aeration tank** in small STPs rated below 5000 m³/day.

This strategy yields a double benefit:

- All downstream units may be placed at a convenient level above ground, resulting in cost savings. At the same time, the maintenance of STP becomes easier.
- The pumping rate can be set at a calibrated uniform flow, so that downstream units are not affected by fluctuating flows.

4.2 How It Works

A typical pair of pumps (working and standby) is shown below:

	S.L	Remarks
	1.	There are two identical pumps . Controls ensure that only one pump can run at a time. Each pump delivers sewage at a rate that is slightly higher than the actual flow rate of the STP.
	2.	Both pumps have independent suction pipes . The inlet pipes extend almost to the bottom of the tank, and must not have foot-valves.
	3.	The delivery pipes from both pumps are combined in a I-shaped header. A delivery pipe takes sewage from this header to the aeration tank.
	4.	The bypass pipeline returns the excess sewage back to the tank.
	5.	Valves fitted on all three pipelines serve different purposes: • The valve on the bypass line is adjusted to “waste” the excess capacity of the working pump. (The delivery pipeline (3) always carries sewage at the designed flow rate) • The valve on the delivery pipe is closed off when the corresponding pump is removed for repairs. This prevents sewage delivered by the other pump from coming out.

4.3 Design Criteria

The capacity of the raw sewage lift pump is selected based on daily average rated capacity of the STP, on the premise that the pumps shall be operated for 20 Hours in a day (For very large STPs, 22 hours of operation in a day may be considered).

STPs are usually designed with a duplicated pumping system: In place of using a single pump, two pumps are fixed in parallel, but only one pump is operated at a time. Such pumps can be operated round the clock (12 hours per pump).

The lifting capacity of the pumps (called 'total head' or 'total lifting height') may be selected based on the level difference between the sewage-delivery level at the aeration tank and the floor level of the equalization tank.

4.4 Construction And Engineering

Despite the presence of the bar screen(s) before the equalization tank, in real-life situations, we cannot rule out the presence of solids, polythene bags, plastic covers, cups etc. in the equalization tank.

These items pose a serious threat to the pumps. Let us compare three different types of pumps for this job:

1. **Submersible pumps** with smaller flow passages in their impellers are not the correct application for this duty: They are prone to frequent failures (either the impeller gets damaged, or the pumps stall and then the winding burns).
2. **Comminatory pumps** with a cutter/shredder option solve the clogging issue by pulverizing the obstacles, but they end up mixing non biodegradable material in the sewage in such a way that separating the material becomes impossible. This is a threat to the environment.
3. Therefore, the correct choice would be horizontal, centrifugal, **non-clog, solids handling (NC-SH)** pumps with open impellers.

There are other valid and practical reasons for this selection:

1. The NC-SH pump is robust for this application and failure rate/ frequency is very low.
2. The NC-SH pumps are rated to handle solids up to even 20 mm size with an open impeller design, whereas submersible pump with closed impeller design comes with smaller openings.
3. The NC-SH pumps are less expensive than submersible pumps, but work at a lower efficiency due to open impeller design. In an STP, robust treatment performance is of prime importance and of higher priority than savings in energy at the cost of treatment efficiency.
4. Repair/ servicing costs for NC-SH pumps are negligible compared to submersible pumps
5. The NC-SH pumps may be serviced at the STP site itself within a few hours with readily available spares and consumables. On the other hand, the submersible pumps have to be sent to their service center/ factory for any repairs, and the time required is typically 2 weeks.
6. Once a submersible pump goes for repair, it never recovers 100% efficiency, and failures start occurring periodically (As per our experience, these pumps are for use and throw duty only)
7. Guarantees/ warranties on repaired units are available, only if sent to the respective factories.
8. The NC-SH pumps are equipped with a Non Return Flap valve in the body itself, which functions as a normal foot valve: hence priming of these pumps is not required at every start.

The raw sewage lift pump is a critical machinery, and so it must have a standby unit. The electrical control circuit must ensure that both pumps cannot run at the same time (otherwise they will generate excessive pressure and damage the plumbing. Also, a higher flow rate means partially treated sewage is passed out of STP.)

Separate suction piping for each of the two pumps is preferred, so that a clogged inlet pipe can be cleaned while the other pump is operating.

The delivery header of the two pumps must conform to good piping engineering practice with necessary fittings for isolating the pumps for maintenance, etc.

It is nearly impossible to get pumps that provide the exact combination of flow rate and head we need. Therefore, a bypass branch line (back to the equalization tank) with a control valve must be provided, so that the sewage flow rate can be precisely set to the designed value.

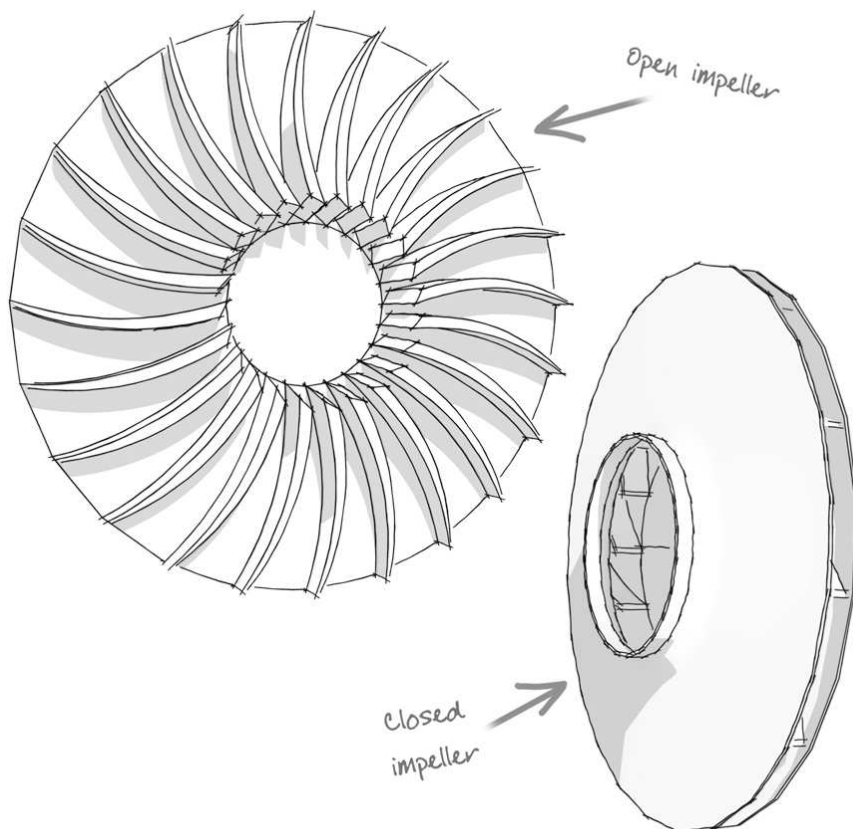
- At the same time, provide for locking this valve, so that the STP operator cannot tamper with its settings to increase the flow rate.

Sufficient space must be allowed around the pump for movement of operators and technicians for routine operation and maintenance activities.

4.5 Operation And Maintenance Considerations

- Switch between the main and standby pump every 4 hours (approximately).
- Check oil in the pump every day; top up if necessary
- Check motor-to-pump alignment after every dismantling operation
- Check condition of coupling and replace damaged parts immediately
- Check for vibrations and tighten the anchor bolts and other fasteners
- Check condition of bearings, oil seals, mechanical seal and replace if necessary
- Completely drain out oil and replace afresh as per manufacturer's recommendation
- Always keep safety guard in its proper Position.
- Ensure discharge of raw sewage into the aeration tank is visible and can be monitored
- Maintain the flow rate at designed level (no tampering with the bypass valve)

Problem	Cause
Excessive noise	Poor engineering/ maintenance
Excessive vibration	Poor engineering/maintenance
Overheating	Poor maintenance
Loss in efficiency of pumping	Poor maintenance



Aeration Tank/MBBR/MBR/SBR/SAFF

5.1 Function

The Aeration tank (together with the settling tank/ clarifier that follows) is at the heart of the treatment system³.

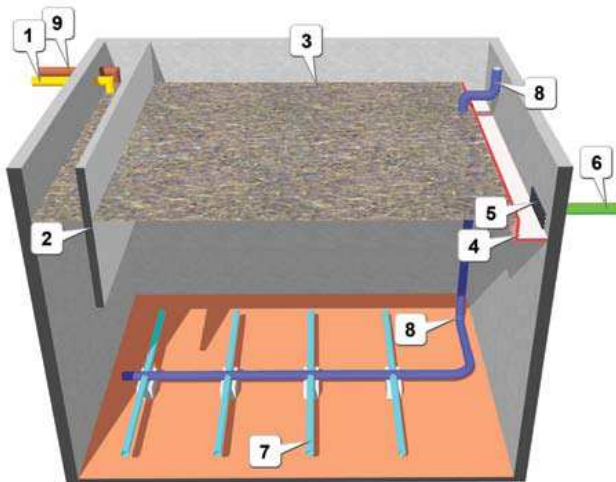
The bulk of the treatment is provided here, employing microbes/bacteria for the process.

The main function of the Aeration tank is to maintain a high population level of microbes. This mixture is called MLSS (Mixed Liquor Suspended Solids).

The mixed liquor is passed on to the clarifier tank, where the microbes are made to settle at the bottom. The settled microbes are recycled back to the aeration tank. Thus they are retained for a long period within the system (see Appendix- page 138).

5.2 How It Works

A typical Aeration tank is shown below.



Notes:

1. The figure shows only the surface of the sewage, so that items submerged in the sewage can be shown.
2. An air-compressor is required, but not shown because in most cases a single blower provides the compressed air needed at multiple places in the STP.
3. Since both tanks work together, they cannot be explained in isolation.
4. This approximation is only to give you a rough idea: The actual ratio depends on the design parameters.
5. The membranes are rated to operate within certain range of air flux rates. So power is saved by turning down (reducing) the air flow during certain times, such as night hours.

SL	Remark
1.	The inlet pipe brings sewage from the raw sewage lift pump (sewage from the equalization tank). The pipe is bent downward, so that the sewage does not get propelled toward the outlet pipe (6).
2.	The baffle wall does not let the incoming sewage and sludge go across the tank toward the outlet pipe (6). The wall forces the mix toward the bottom of the tank; thus ensuring maximum retention.
3.	The tank is always filled till this level (which is set by the top of the launder (4) . So the remaining height of the tank serves as freeboard (height margin to ensure that the tank does not overflow immediately under moderate emergencies.)
4.	The Outlet Launder collects the sewage and delivers it to the outlet pipe(6). Note that the outlet launder is located farthest from the inlet pipe (1) to minimize short circuiting of flow from the inlet to the outlet of the tank.
5.	The net prevents entry of debris in the outlet pipe (6). The operator should remove debris collected in the launder (4) periodically, otherwise eventually the mesh will be blocked with accumulated debris, resulting in a rise of water-level in the aeration tank. In the extreme case, this will cause overflow from the tank.
6.	The outlet pipe takes the sewage to the settling tank/secondary clarifier.
7.	The fine bubble diffusers are actually rigid pipes with long slots, which are then covered with tubular synthetic rubber membranes. The compressed air is released in the form of fine bubbles throughout the length of the diffusers, through minute holes punched in the rubber membrane.

	The figure shows an array of eight diffusers. The array is strapped to cement blocks (ballasts) to keep the entire assembly anchored to the bottom of the tank.
8.	In the case of fixed diffusers, compressed air is supplied through a header pipe at the bottom of the tank, as shown. Some designs use flexible air hose lines and pairs of diffusers to make them easily retrievable. In this case each pair of diffusers is also provided with a nylon rope to enable lifting out of the aeration tank for maintenance.
9.	The recalculated sludge pipeline brings bacteria floc from the settling tank/secondary clarifier). It is always located very close to the inlet pipe (1) so that the raw sewage and bacteria get mixed thoroughly. (By design, both pipes deliver roughly the same volume per hour ⁴ .)

5.3 Design Criteria

When designing an STP, the following factors are already known:

- The quantity of sewage to be handled per day
- The pollution potential of Indian domestic sewage, in terms of commonly understood parameters such as BOD (Biochemical Oxygen demand), COD (Chemical Oxygen Demand), TSS (Total Suspended Solids), O&G (Oil & Grease), etc.

Thus we know the amount of food available every day for the microbes to eat away.

The other factors are selected as follows:

Treatment efficiency	90 to 98 %, as defined by the Pollution Control Board.
Food/ Microorganisms ratio (F/M)	For STPs with extended aeration, required F/M is 0.10 to 0.12

This gives the required size (volume) of the aeration tank.

The next step is to calculate the amount of air to be pumped into the aeration tank, to keep the microbes alive and in continuous suspension (they must mix well with the food, and not settle at the bottom of the tank).

- In fact, the amount of air required for respiration of the microbes is always more than the amount of air required to keep the tank contents completely mixed. Therefore, we can simply calculate the air required for microbes; and it will serve the other purpose well.

The thumb rule is 50-60 m³/hr of air for every kg of BOD removed (i.e., the difference between BOD readings of the incoming sewage and treated sewage).

That concludes the design of the Aeration tank: the size (volume), concentration of microbes to be maintained, and the quantity of air to be supplied per hour.

5.4 Construction And Engineering

The Aeration tank is generally of waterproof RCC construction (as are most other tanks in the STP), designed as water-retaining structures as specified in relevant Indian codes.

The shape of the tank is not very critical, as long as adequate floor coverage and uniform mixing can be achieved by proper placement of diffusers on the tank floor.

Operating platforms must be provided next to the tank, such that all the diffusers installed in the tank are easily accessible, and amenable to easy maintenance.

In theory, the desired volume can be achieved with multiple combinations of tank dimensions. However, in practice, the following factors limit the depth of the tank:

- The sewage depth may be between 2.5 - 4.0 m. The greater the water depth, the higher the efficiency of transfer of Oxygen to the tank contents. However, there is a penalty to be paid in the form of higher (and more difficult maintenance, costs of a higher pressure air blower, higher air temperatures and related problems.
- Requirement for headroom above the tank, for operator comfort and to allow maintenance (e.g. to retrieve the heavy diffusers from bottom, you may need to fix a pulley system on the ceiling)

So the depth is fixed first. The length and width of the aeration tank may be computed to suit the diffuser

membranes selected to provide the required quantity of air.

It is best to use the least possible number of membranes and therefore use the largest of the available sizes: 90 Dia x 1000 mm long. The lesser the number of membranes, the lesser is the maintenance, and the fewer the chances of malfunction.

Membrane diffusers are the preferred equipment for aeration in the aeration tank over other forms of aeration (low-speed surface aerators/ High-speed floating aerators/ submerged venture aerators, etc.) for several reasons:

- Energy savings
- Less number of rotating machinery to be operated and maintained
- Turndown option⁵
- Standby facility
- Gentle aeration (less breakage of the biomass floc)
- Performance unaffected by foaming in tank
- Substantial reduction in aerosol formation (Safe working conditions)
- Mixing in depth
- Design can take into account the following factors: pressure, temperature, altitude, viscosity, fouling, aging, etc.
- Non-interruptive maintenance/ replacement is possible

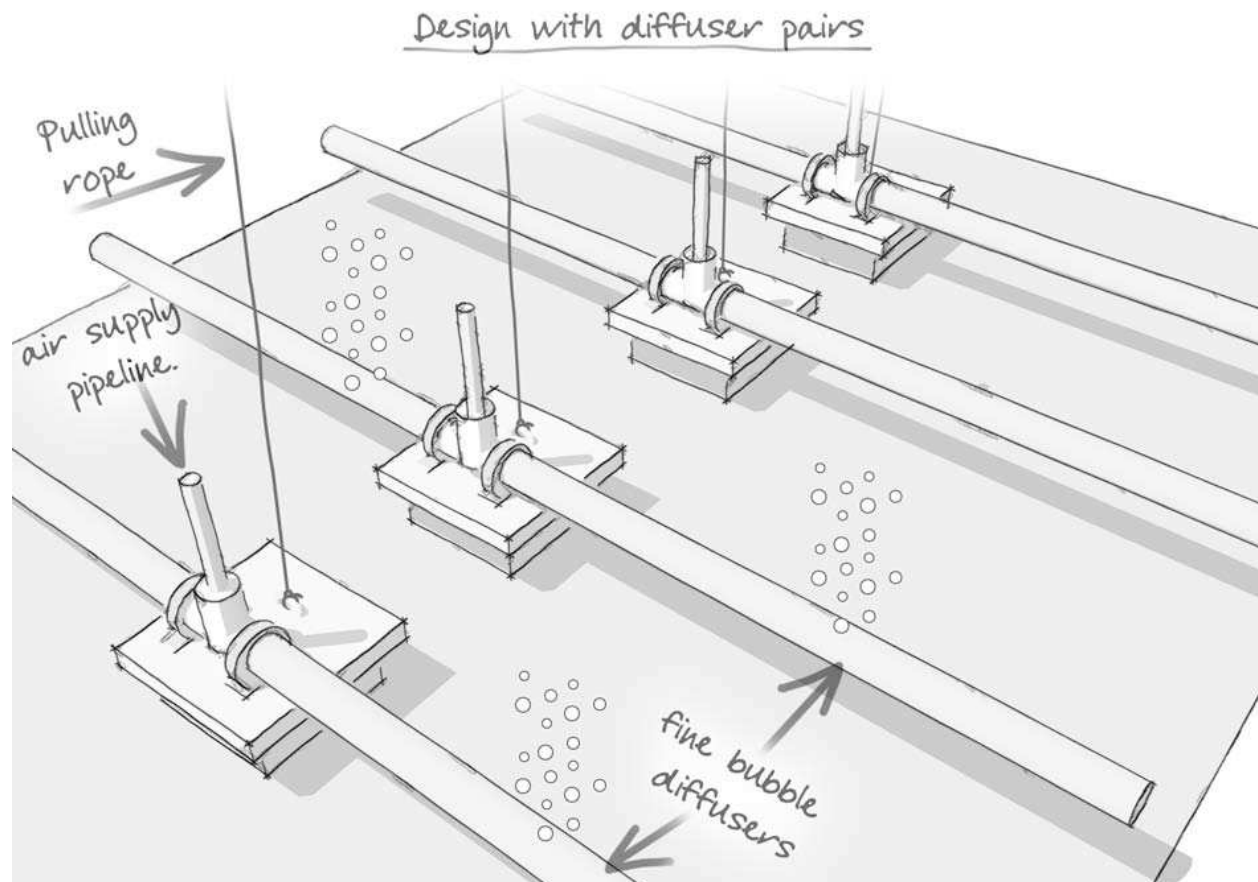
The diffusers must be retrievable, for regular cleaning and maintenance without having to empty the aeration tank. (Regular cleaning extends the life of the diffusers). It is necessary to ensure that the incoming sewage does not go to exit directly. To minimize this "short circuiting", raw sewage lift pumps must deliver the sewage at one end of the tank, and the outflow must be as far away from this point. For the same reason, the sludge recirculation pipe (from the settling tank) must deliver sludge in the vicinity of the sewage inlet, to maximize the contact time of microorganisms with raw sewage. The outlet end may be provided with a launder at the desired water level in the tank (which in fact fixes the water level in the tank). It is also useful to fix a coarse mesh screen in the launder to trap any stray trash from entering the secondary settler tank. Sufficient freeboard must be provided in the tank, so that even in the event of emergencies (such as blockage of pipe between aeration tank and settling tank, excessive foaming etc.) overflow from the aeration tank can be avoided for some time. Note that the freeboard only gives the STP operator some additional time to react to an emergency, but it would not be able to prevent an over flow. All things considered, chances of poor engineering in the aeration tank affecting STP performance are far less compared to the settling tank (secondary clarifier-- the next tank in the chain).

5.5 Operation And Maintenance Considerations

Operation considerations include maintaining the correct design level of MLSS (biomass concentration) in the aeration tank. Problems arise both in the case of excess or shortage of biomass, causing an imbalance, leading to failure of the process. The next chapter shows how to maintain the correct design level of MLSS in the aeration tank. See appendix (page 139) to understand how MLSS ratio is measured and controlled. Visual observation will indicate if there is uniform aeration and mixing over the entire area of the tank. Local violent boiling/ bubbling is indicative of ruptured membranes. Dead zones on the sewage surface indicate that membranes are blocked from the air side or the liquid side. Both conditions call for immediate attention, by cleaning or replacing the membranes. Cleaning of membranes is generally carried out by lifting out the defective units and scouring out the adhering materials by high-pressure hosing. Scrubbing with mild acid solution may also be resorted to in case of stubborn encrustation. Foaming in the aeration tank may be caused by excessive inflow of detergent-like substances: In a great majority of cases, the cause may be traced to an imbalance in the aeration tank recipe (Food Microorganisms: Air: Nutrients), and corrective measures may be taken as indicated.

5.6 Troubleshooting

Problem	Cause
Inadequate mixing/ aeration	Poor design/ engineering/ maintenance
Violent boiling in tank	Ruptured membranes/ damaged pipeline
Black coloration (medium to dark brown color indicates good health)	Poor design/ engineering
Foaming Note: Foaming during initial start-up of STP is normal, due to the acclimatization period of the bacteria in the growth phase.	Poor design/ engineering/ operation
Paucity of bacteria • Very light colored liquid in Aeration Tank • MLSS below acceptable limits	Poor design/ engineering/ operation



Secondary Clarifier/ Settling Tank

6.1 Function

The purpose and function of the secondary clarifier is threefold:

- Allow settling of biomass solids in the Mixed Liquor (biomass slurry) coming out of the aeration tank, to the bottom of the clarifier
- To thicken the settled biomass, in order to produce a thick underflow
- To produce clear supernatant water, in the overflow from the clarifier
- The clarifier tank is only a passive device: All the above actions occur due to gravity.

The thick biomass is recalcitrated back to the aeration tank.

6.2 How It Works

The clarifier tanks can be classified in two groups: mechanized and unmechanized. In an unmechanized clarifier, the bottom of the tank is shaped like a funnel, with a steep slope. The sludge slowly settles towards bottom, and slides down the slope to collect at the lowest point of the funnel-shaped bottom. In a mechanical clarifier, the bottom of the tank has only a gentle slope toward the center. The sludge settles uniformly across the floor of the tank. A set of slowly rotating rubber blades sweep the sludge into a hopper at the center of the tank⁶.

There are three popular design variations in the unmechanized clarifier tank. They differ in the manner in which the sludge is collected and returned to the Aeration tank.

Electric pump- Direct suction	Sludge is collected with an electric pump. Since the flow rate of this pump cannot be varied, the pump is turned off periodically to arrive at a lower net flow rate. (For example, it is kept off for 10 minutes every hour.)
Electric pump and buffer sump	Sludge is allowed to flow in a buffer sump (using gravity). From here, sludge is pumped back to aeration tank using a pump. The net flow rate is adjusted using a bypass pipeline with a valve (exactly like the raw sewage lift pumps).

The unmechanized and mechanical varieties of clarifiers are explained next.

6.2.1 Settling tank with air-lift pump

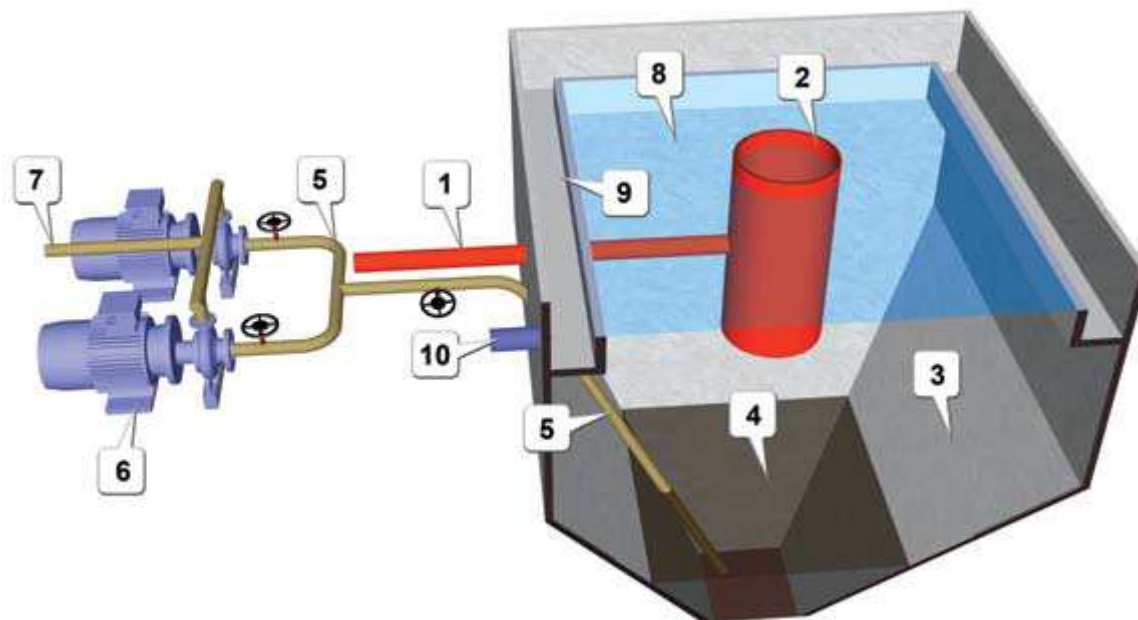
A typical settling tank with air-lift pump is shown on the next page.. (The front side is removed to show internal parts.)

S.L	Remark
1.	The sewage inlet pipe brings sewage from the aeration tank.
2.	The center-feed well (also called "influent well") takes this incoming sewage and gently releases it in the settling tank, without causing any disturbance or turbulence. Note that the well is always filled with water because of its position. So the incoming sewage does not drop from a height and disturb the sludge that is already settling toward the bottom of the tank. Also note that the top of the well is positioned above the water surface, so that the incoming water cannot find a path of least resistance, straightway rise to the top and exit to the launder. (If that is allowed to happen, then the solids will never be able to settle.)
3.	The sludge is only slightly heavier than water; so it takes time to sink. It slides down the steeply sloped walls of the tank toward the center of the bottom.
4.	The bacterial flocs ⁷ collect here in high concentrations ⁸ . Even when the flocs settle at bottom, they actually remain suspended in water, rather than forming a solid sediment. Note: This figures shows it as semitransparent only to show the suction pump mechanism. In real life, the mix would be a dense opaque mass. The upper part of the tank, till the surface, holds clear water (the figure shows only its surface ⁸).

5.	The compressed air pipe (5) feeds compressed air to the airlift pump. As this air is released near the bottom, it expands suddenly in bubble form and rises to the top through the delivery pipe (7). The Venturi effect at the joint of 5 and 7 creates a partial vacuum, and sucks in the bacterial flocs through the inverted funnel (6). The compressed air propels this mass up through the delivery pipe (7).
6.	8 The clear water rises to the top of the tank.
7.	9 Typically the tank has launders on all four sides. (The launder on the front is not shown, so that the other parts can be shown clearly.) If the water overflowed over the weir at fast velocity, it will pull up the solids from the bottom of the tank. This is prevented by providing weir of sufficient length. In small plants, launder on a single side is adequate.
8.	10 The clarified water pipe takes the decanted water to the clarified water sump.

6.2.2 Settling tank with direct suction electric pump

A typical settling tank with a direct-suction electric pump is shown below. (The front side is removed to show its internal parts.)



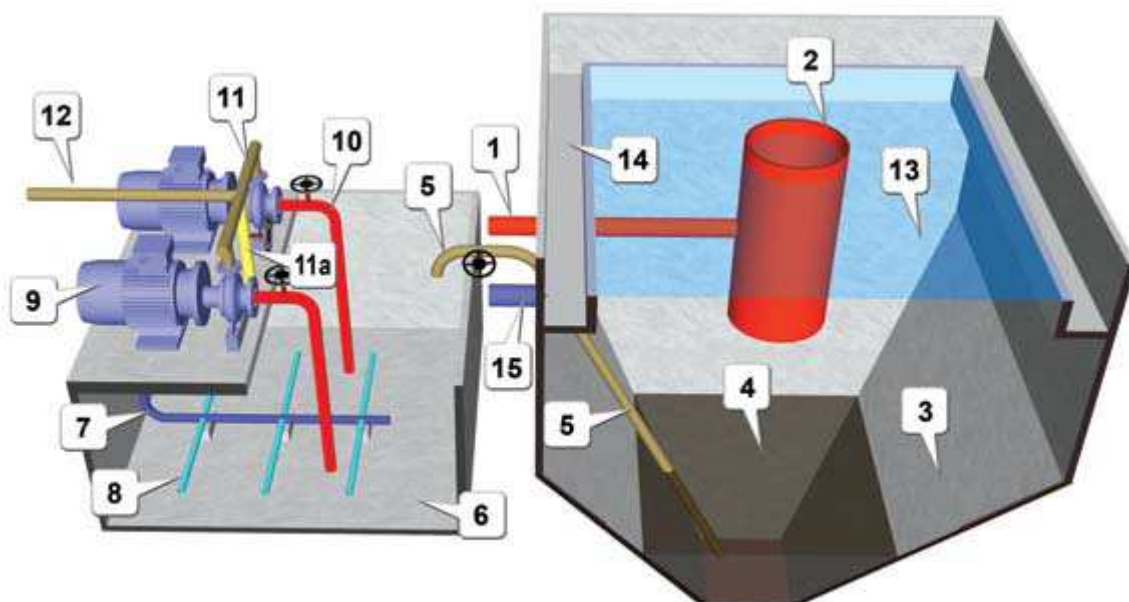
SL	Remark
1.	The sewage inlet pipe brings sewage from the aeration tank.
2.	The center-feed well (also known as "influent well") takes this incoming sewage and gently releases it in the settling tank, without causing any disturbance or turbulence. Note that the well is always filled with water because of its position. So the incoming sewage does not drop from a height and disturb the sludge that is already settling toward the bottom of the tank. Also note that the top of the well is positioned above the water surface, so that the incoming water cannot find a path of least resistance, straightway rise to the top and exit to the launder. (If that is allowed to happen, then the solids will never be able to settle.)
3.	The sludge is only slightly heavier than water; so it takes time to sink. It slides down the steeply sloped walls of the tank toward the center of the bottom.
4.	The bacterial flocs ₉ collect here in high concentration ₁₀ . Even when the flocs settle at bottom, they actually remain suspended in water, rather than forming a solid sediment. Note: This figures shows it as semitransparent only to show the suction pump mechanism. In real life, the mix would be a dense opaque mass. The upper part of the tank, till the surface, holds clear water (the figure shows only its surface (8).
5.	The sludge delivery pipe delivers ₁₁ the slurry (a mix of flocs and water) to the pumps (6). The pipe is split between the two pumps (they do not have separate inlet pipes). The valve on the main pipe is used to close it during maintenance. The valves near the pumps regulate the

	flow, and also close the pipe when the corresponding pump is removed for repairs.
6.	There are two identical pumps. Controls ensure that only one pump can run at a time. Each pump delivers sludge at a rate that is slightly higher than the required flow rate. Since the flow rate of these pumps is fixed, they need to be turned off periodically to bring down the net flow rate to achieve the desired MLSS ratio. This is a critical operation, because if flocs remain in the settling tank for more than 30 minutes, the microorganisms die due to lack of oxygen. Therefore the on/off cycles have to be small. Since this is a round-the-clock operation, and a critical task, this is done using a timer circuit that turns the pump on/ off automatically. The operator has to monitor the MLSS ratio, and keep adjusting the timer's duty cycle (the on/ off periods) as required.
7.	The I-shaped header assembly joins the outlet pipes of both pumps, and delivers the sludge to the aeration tank.
8.	The clear water rises to the top of the tank.
9.	Typically the tank has launders on all four sides. (The launder on the front side is not shown, so that the other parts can be shown clearly.) If the water overflowed over the weir at fast velocity, it will pull up the solids from the bottom of the tank. This is prevented by providing weir of sufficient length. In small plants, launder on a single side is adequate.
10.	The clarified water pipe takes the decanted water to the clarified water sump.

Because of the requirement to switch the pump on/off round the clock (and the consequences of an operator mistake), this method is not recommended. The next variant overcomes the design limitations mentioned above.

6.2.3 Settling tank with buffer sump

This design overcomes the critical deficiencies in the direct-suction electric pump version. The settling tank is built just like the direct-suction electric pump version, but two critical external components are added: An intermediate tank with aeration, and a bypass pipeline. These additional components keep the bacteria flocs alive, and also allow the operator to adjust the sludge flow rate to get the desired MLSS ratio. A typical settling tank with buffer sump is shown below. (The front side is removed to show its internal parts.)



SL	Remark
1.	The sewage inlet pipe brings sewage from the aeration tank.
2.	The center-feed well (also known as "influent well") takes this incoming sewage and gently releases it in the settling tank, without causing any disturbance or turbulence. Note that the well is always filled with water because of its position. So the incoming sewage does not drop from a height and disturb the sludge that is already settling toward the bottom of the tank. Also note that the top of the well is positioned above the water surface, so

	that the incoming water cannot find a path of least resistance, straightway rise to the top and exit to the launder. (If that is allowed to happen, then the solids will never be able to settle.)
3.	The sludge is only slightly heavier than water; so it takes time to sink. It slides down the steeply sloped walls of the tank toward the center of the bottom.
4	The bacterial flocs ¹² collect here in high concentration ¹³ . Even when the flocs settle at bottom, they actually remain suspended in water, rather than forming a solid sediment. Note: This figures shows it as semitransparent only to show the suction pump mechanism. In real life, the mix would be a dense opaque mass. The upper part of the tank, till the surface, holds clear water (the figure shows only its surface 8).
5.	The sludge delivery pipe delivers ¹⁴ the slurry (a mix of flocs and water) to the buffer sump (6). Note that the pipe is located below the water surface, therefore the slurry is delivered with pressure. The valve on the pipe is used to regulate the slurry flow rate and also to close the pipe during maintenance.
6.	The slurry remains in the sump (6) only for a short time. The compressed air pipeline (7) provides air to the coarse air bubble diffusers (8), which release large bubbles in the slurry. This not only provides oxygen to the bacteria, but also continuously agitates the slurry to prevent settling of the flocs to the bottom of the sump ¹⁵ .
7.	There are two identical pumps which pass the slurry to the Aeration tank. Controls ensure that only one pump can run at a time. Each pump delivers sludge at a rate that is slightly higher than the required flow rate. The extra flow is diverted back to the sump through the bypass pipeline (10). The returning slurry is released at a height, thus agitating the contents of the sump. The valve on the bypass line is adjusted to achieve the desired net flow rate.
8.	The !-shaped header assembly joins the outlet pipes of both pumps, and delivers the sludge to the aeration tank via the delivery pipe (12).
9.	The clear water rises to the top of the tank. (This figure shows only the surface, but the upper part of the tank is filled with clear water).
10	Typically the tank has launders on all four sides. (The launder on the front side is not shown, so that the other parts can be shown clearly.) If the water overflowed over the weir at fast velocity, it will pull up the solids from the bottom of the tank. This is prevented by providing weir of sufficient length. In small plants, launder on a single side is adequate.
11	The clarified water pipe takes the decanted water to the clarified water sump

Each floc is loosely aggregated mass of bacteria. It is a brownish tiny ball of 2-3 mm dia, with a soft, spongy and slimy texture.

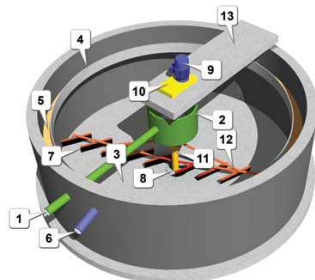
This simplified diagram shows a separate layer of flocs at bottom. In reality, at any given moment, the newly arriving flocs are gradually sinking, and clear water is rising upward. This creates a gradual increase of floc density toward the bottom of the tank, but there are no distinct layers.

Because of the water column above, the slurry is delivered with pressure. Thus the pumps do not need to apply suction: They work only to lift the slurry to the top of the aeration tank.

6.2.4 Mechanized Clarifier Tank

The three types of clarifier tanks described so far were not mechanized: In those tanks, the sludge settles and moves to the deepest part of the tank due to gravity, from where a pump takes it to the aeration tank. In a mechanized clarifier tank, the sludge settles at the bottom over a wide area, and a few rubber wiper blades (called "squeegees") sweep it to a pit at the center of the tank, from where a pump takes it to the aeration tank.

This design is used for large STPs only. A typical tank is shown below. As shown, the tank is cylindrical, with bottom that slopes toward the center, with very little slope. The figure does not show the clarified water and sludge, so that the submerged parts can be shown clearly,



The sewage inlet pipe (1) brings sewage from the aeration tank. The center-feed well (also known as “influent well”) (2) takes this incoming sewage and gently releases it in the settling tank, without causing any disturbance or turbulence. Note that the well is always filled with water because of its position. So the incoming sewage does not drop from a height and disturb the sludge that is already settling toward the bottom of the tank. Also note that the top of the well is positioned above the water surface, so that the incoming water cannot find a path of least resistance, straightway rise to the top and exit to the launder. (If that is allowed to happen, then no settling of solids will occur.) The sludge is only slightly heavier than water; so it takes time to sink. The bacterial flocs¹⁶ collect at the bottom in high concentration¹⁷. Even when the flocs settle at bottom, they actually remain suspended in water, rather than forming a solid sediment. Since the bottom (3) has very little slope, the flocs do not move from their settling spot. The clear water rises to the top. Most of the tank is filled with clear water (not shown here). As we go down, we find the sludge in progressively higher concentration. The tank has a launder (5) around its periphery. The clear water flows over its top edge (4) (called “weir”) into the launder, and is collected by the outlet pipe (6) and taken to the filter units (pressure sand filter and activated carbon filter). Note that the position of the launder determines the depth of water in the tank: Under normal running condition, the water level never rises beyond the weir. It rises only when the outlet pipe is blocked for some reason. So the remaining height of the tank above the weir serves as freeboard (height margin to ensure that the tank does not overflow immediately under moderate emergencies.) The bacterial flocs settled at the bottom have to be quickly collected and sent back to aeration tank, otherwise they would die because of lack of oxygen and food. This is done with a set of rubber blades (called “squeegees”) (7): The squeegees sweep the floor in circular movement, and propel the sludge toward a collection-pit (8) at the center of the tank. From here, a pump takes the sludge to aeration tank. The squeegees work just like how a windshield wiper works in your car. Note how each squeegee is set at an angle. When a squeegee sweeps through the layer of accumulated sludge, the sludge slides towards its trailing edge. The squeegee leaves a continuous ridge of sludge at the trailing edge. (The five squeegees on each arm leave five ridges of sludge behind.)

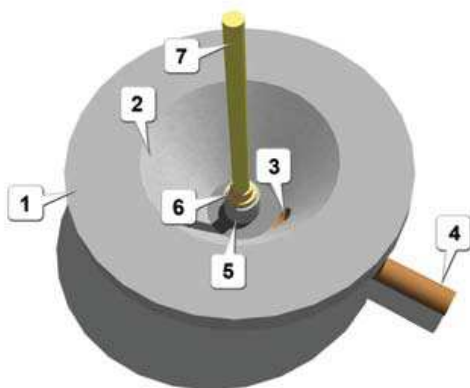
Each floc is loosely aggregated mass of bacteria. It is a brownish tiny ball of 2-3 mm dia, with a soft, spongy and slimy texture.

This simplified diagram shows a separate layer of flocs at bottom. In reality, at any given moment, the newly

Each ridge is then collected by the next squeegee in the opposite arm of the rake (after the rake turns by half a turn), and again pushed toward the center. Thus each pass of the squeegees moves the sludge toward the center by one blade length. The last squeegees (nearest the center) drop the sludge into the collection pit. Thus, if each arm has five squeegees, the sludge will take up to five rotations of the rake to reach the pit. Also note that the squeegees are arranged to cover the entire floor of the tank; especially the outer periphery. This is a critical requirement: It ensures that the bacterial flocs do not remain in the Clarifier tank for extended period and die. The circular motion of the squeegees is achieved through the following mechanism: A motor (9) drives a speed-reduction gear box (10). The gear box drives a shaft (11). The shaft is attached to the frame (12) that carries all the squeegees (7).

Thus when the motor rotates at its normal speed, the frame rotates at a very slow speed in the tank. This slow speed ensures that the sludge settled at the bottom of the tank is not stirred up. The platform (13) (also called “bridge”, because it spans across the tank in most designs) ¹⁸ allows mounting of the driver motor, and also allows the STP operator to observe the tank from above. It always has safety hand-rails (not shown in the figure to reduce complexity.)

Some critical details of the sludge-collection pit are shown below:



The figure shows only some part of the tank floor (1). A bucket-shaped sludge-collection pit (2) collects the sludge that is swept by the squeegees. The collected sludge is pumped out through the outlet port (3) and outlet pipe (4). In the center of the pit, an RCC pillar (5) is provided. A bush housing (6) is mounted on this pillar. The housing contains a bush, which provides a frictionless support to the rotating rake (7). This ensures that the rotating rake remains steadily centered in the tank; and more importantly, the squeegees remain in contact with the floor as they rotate. If a bush is not provided, the rake would be dangling from the platform. As it rotates in the thick slurry, it meets uneven resistance and currents; and starts swinging. As a result, the squeegees cannot remain in touch with the floor, and their sweeping action is not uniform. That leaves a lot of sludge unswept on the floor. A second major problem is the rubber squeegees strike the floor violently, and get damaged. Such torn squeegees cannot sweep the sludge properly. Again, this leaves a lot of sludge unswept on the floor. Bacteria that cannot be collected within one hour from the clarifier tank die, and turn septic. Thus the bottom bush plays a vital role. In some designs, a sludge-concentrator (not shown here) is fitted on the central shaft (7). The concentrator is like a large screw. It rotates with the rake, and pushes/squeezes the flocs down the pit. At the bottom of the pit, the floc concentration is the highest. This increases the density of the slurry that is returned to the aeration tank.

6.3 Design Criteria

The fundamental design criterion for clarification of the mixed liquor coming into the clarifier is the cross-sectional area of the clarifier. In the strictest theoretical sense, the depth of the clarifier has no role to play in the "clarification" function: Increasing depth of the clarifier only helps in the "thickening" function. Clarifier cross-sectional area is typically computed at between 12 – 18 m³/hr/m² of throughput flow of sewage, depending on various other factors. For small domestic STPs, a figure of 16 may be taken as the golden mean. It is customary to specify depth of clarifier between 2.5 to 3.0 m. In order to restrict localized high upflow velocities, clarifiers have to be provided with sufficient length of "weir" over which overflow occurs. In small clarifiers, the "Weir Overflow Rate" does not assume critical significance, and in a square tank, a weir on a single side of the tank will be sufficient. Circular clarifiers in most cases are built with an all-round weir, which again is adequate. Another design consideration is the "Solids Loading Rate" in the clarifier - i.e. kg solids/m²/Hr. In typical domestic STPs, this parameter is not of great significance.

6.4 Construction And Engineering

Proper construction and engineering of a clarifier/ settling tank is of utmost importance, and several factors need to be considered and executed with great precision. Any deficiency in even one of these aspects can make or mar an STP. Some of the more critical factors are listed below:

- Steep slope in the hopper-bottom settling tank (> 45°)
- Weir at uniform level
- Influent feed well to kill turbulence of incoming mixed liquor from the aeration tank
- Radial entry of mixed liquor into the feed well
- Minimum difference in water level in aeration tank and clarifier (not more than 0.2 m)
- Minimum footprint of the central sludge collection hopper
- If a square tank is fitted with a mechanical rake, its corners (which are not swept by the blades) must have steep slope.
- Uniform slope in floor, without major undulations
- Rubber squeegees of the mechanical rake to sweep the floor
- Number, angle and length of rake blades on the rake arm
- Speed of the rake arm
- Bottom "steady bush" for the vertical shaft of rake arm in large clarifiers to prevent oscillation of the rake arms
- If the sludge-withdrawal pipe is buried beneath the floor in a mechanical clarifier, it shall be minimum 4" diameter.

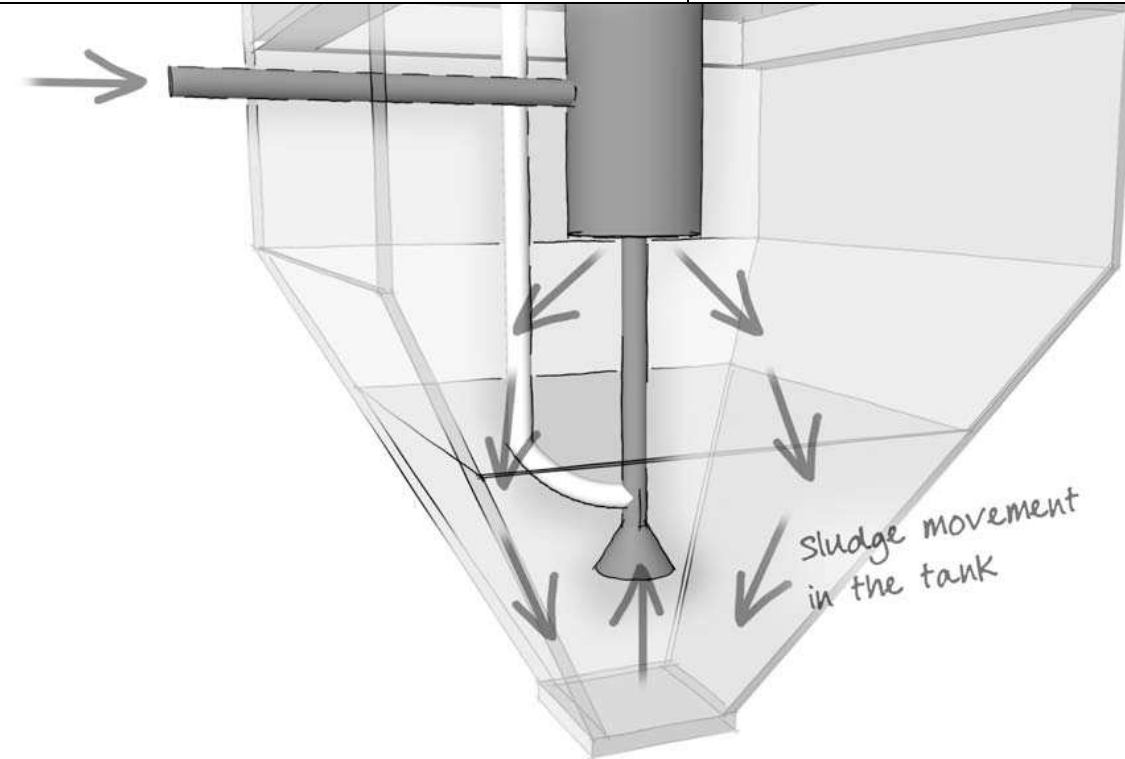
This figure shows only half of the bridge, so that the items located under it can be shown clearly. *Secondary Clarifier/ Settling Tank*

6.5 Operation And Maintenance Considerations

If properly designed, engineered and constructed, clarifiers call for very little attention in terms of operation and maintenance. Indeed, the unmechanized (hopper-bottom) settling tanks may be said to be zero- maintenance units. Some parts of the mechanical rake (such as the motor, gearbox etc.) call for only routine maintenance. The sacrificial rubber squeegees sweeping the floor of the clarifier need to be checked and replaced, possibly once in two years. *Secondary Clarifier/ Settling Tank*

6.6 Troubleshooting

Problem	Cause
Solids are carried over with decanted water	Poor design/ engineering/ operation/ maintenance
Mushroom cloud of solids	Poor engineering/ maintenance
Poor settlement of solids	Poor design/ engineering
Thin slurry in underflow	Poor design/ engineering
Excessive turbulence in clarifier	Poor engineering
Rotational flow of solids in upper layers ¹⁹	Poor engineering



It fact, any movement of water in the clarifier should not be noticeable at all. Expert near the overflow wears where the velocities are high.

Sludge Recirculation

7.1 Function

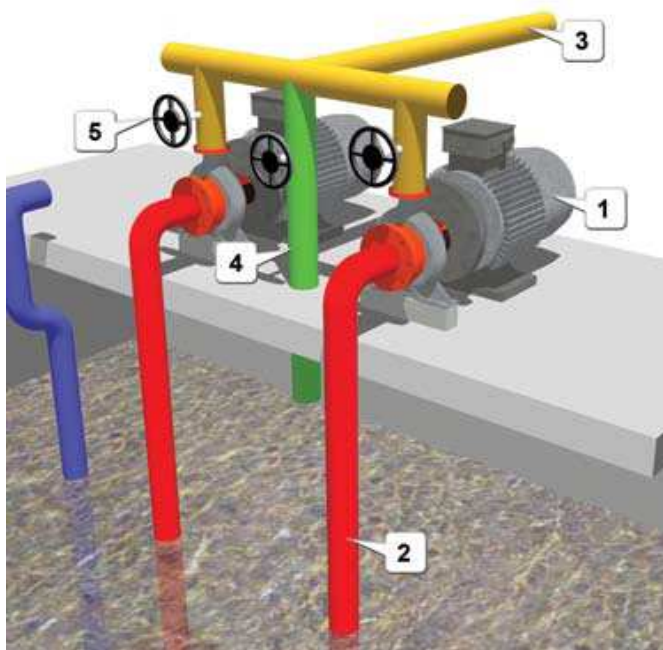
The indivisible combination of the aeration tank, settling tank and sludge recirculation constitutes an “activated sludge biological treatment system”. All three must be fine-tuned to act in unison to produce the desired high level of treatment. The optimum desired age of the microbes is between 25 to 30 days. At the same time, an STP

7.2 How It Works

A typical pair of pumps (working and standby) is shown below:

Sludge Recirculation needs to maintain a high level of microbes in the aeration tank. Both these objectives are achieved by recirculating the sludge from the settling tank, and also bleeding out of excess microbes from the system at regular intervals. To illustrate by a simple example: if the total biomass inventory in the system is 100 kg, and daily bleed/ wasting rate is 4 kg, then the average age of biomass in the system is 25 days.

Note: The example shows the pipelines in different colors only for illustration purposes. In actual practice, no such color-coding is followed.



SL	Remark
1.	There are two identical pumps. Controls ensure that only one pump can run at a time. The sludge from the sludge hopper of the clarifier is taken by gravity into a sludge sump, from where the pumps return the sludge to the aeration tank.
2.	Both pumps have independent suction pipes - It is not desirable to have a common suction pipeline, because if it fouls up, both pumps will have to be shutdown. - The pipes must not have foot-valves, because the foot-valves would get jammed frequently. - The inlet pipes extend almost to the bottom of the tank
3.	The delivery pipes from both pumps are combined in a I-shaped header. A delivery pipe takes sludge from this header to the aeration tank
4.	The bypass pipeline returns some sludge back to the sludge-holding tank.
5.	5Valves fitted on all three pipelines serve different purposes :The valve on the bypass line is adjusted to “waste” the excesscapacity of the working pump. (The delivery pipeline (3) always carries sludge at the designed flow rate, to achieve the desired MLSS in the aeration tank.) - The valve on the delivery pipe is closed off when the corresponding pump is removed for repairs. This prevents sludge coming from the other pump from coming out.

7.3 Design Criteria

Sludge recirculation rates are typically between 50 % to 100 % of the throughput rate of sewage in the STP. Hence, in a majority of cases, the capacity and specifications of the raw sewage lift pumps are replicated for this duty as well.

7.4 Construction And Engineering

The engineering principles prescribed for the raw sewage lift pumps (see section 4.3) apply to the sludge recirculation pumps as well. Providing an intermediate sludge sump (between the clarifier and the recirculation pumps) to collect sludge from the bottom of the clarifier tank is preferable to directly connecting the pump to the sludge pipe of the clarifier: This strategy enables control of the recirculation sludge rate, without having to throttle the pump, thereby reducing pump-maintenance costs and extending life of the pumps. Typically in small plants (say up to 150 m³/day) 5-10 m³/hr of air is sufficient to lift sludge in a 2”-3” dia sludge pipe, and deliver it back to the aeration tank. Note that air-lift pumps work well only when the submergence is high (i.e., when the mouth of the airlift pump is deep inside the sewage) and the delivery head is small (i.e., when the Aeration tank’s top is not much above the top level

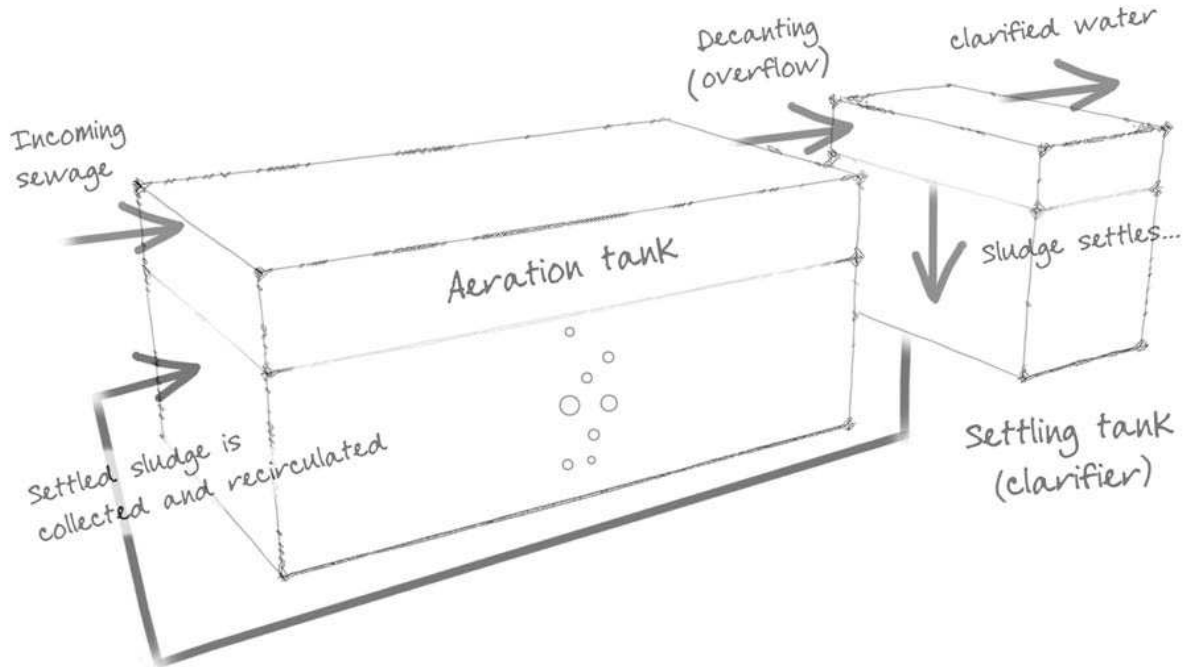
of the settling tank). In other words, the water level difference between the Aeration Tank and Settling Tank must not be excessive; otherwise the pump will not be able to lift the sludge, and thus the recirculation will stop.

7.5 Operation And Maintenance Considerations

Considerations here are identical to those specified for the raw sewage lift pumps (see Section 4.4). The manufacturer's O&M manual must be followed with diligence. Ensure discharge of sludge recirculation into the aeration tank is visible and can be monitored. In addition, if an intermediate sludge sump is provided, it is advisable to force-flush the sludge line of the clarifier at frequent intervals, so that the pipe remains clear at all times, and incidence of choking is minimized. *Clarified Water Sump*

7.6 Troubleshooting

Problem	Cause
Excessive noise	Poor engineering/maintenance
Excessive vibration	Poor engineering/ maintenance
Overheating	Poor maintenance
Loss in efficiency of pumping	Poor maintenance



Clarified Water Sump

8.1 Function

Overflow water from the clarifier is collected in an intermediate clarified water sump. This sump acts as a buffer tank between the secondary and the tertiary treatment stages in an STP. In a well-run STP, the treated water quality at this stage is good enough for reuse on lawns and gardens with sufficient disinfection, and water for garden use may be directly taken from this sump, without having to overload the tertiary units. Also, during lean inflow periods to the STP, backwashing of the filters is carried out. At this time, this tank must hold sufficient buffer stock of water for backwash purposes.

8.2 Design Criteria

Any sump tank that serves pumps should have a minimum retention period of 30 minutes, so that only under extreme negligent operations, the sump may overflow, or the pump may run dry. In addition, the tank must hold enough water to back flush the filters fully. Thus it is prudent to provide a retention time of 2-3 hours of average hourly flow in the STP. Despite best design, trace quantities of solids always escape the clarifier into this tank. This means presence of live bacteria in this tank. Therefore, it is advisable to aerate this tank, in order to keep the bacteria alive and keep the water fresh. The air bubbles also serve another purpose: The compressed air keeps these solids in continuous suspension by constantly agitating the water. This prevents the solids from settling at the bottom of the sump and accumulate there. (Settled bacteria will eventually starve and die, as this tank does not have enough food for them. That would turn the contents of the tank septic.)

8.3 Construction And Engineering

The tank should be able to properly feed the suction pipeline of the filter feed pumps. Minimum aeration with coarse bubble diffusers is recommended in this tank to prevent settling of the trace amounts of suspended solids slipping through the settling tank. It should be possible to clean and maintain the diffusers with ease.

8.4 Operation And Maintenance Considerations

There are no special requirements, as this tank plays a passive role in STP functioning. In general, look after aeration, and inspect the tank periodically for sediments. Remove sediments as required.

8.5 Troubleshooting

Troubleshooting in this unit of the STP is not called for due to its passive role.

Filter Feed Pumps (FFP)

9.1 Function

Filter feed pumps are used to take the water from the clarified water sump and pass it through the pressure sand filter and activated carbon filter installed in series.

9.2 Design Criteria

Capacity of the filter feed pumps may be chosen keeping in view the desired number of hours of operation of filters (if not the standard 20-22 hours of operation). In this case, the capacity of the intermediate clarified water sump also needs to be enhanced accordingly. The discharge head of the filters may be specified at 1.5 - 2 kg/cm², to overcome the pressure differential across the two filters under the worst condition (which is just before backwashing or back flushing the filters).

9.3 Construction And Engineering

The filter feed pumps may be selected either to be of the open impeller type (more efficient) or, we may fall back upon the trusted non-clog, solids-handling (NC-SH) type of pump selected for raw sewage lifting (see Section 4.4).

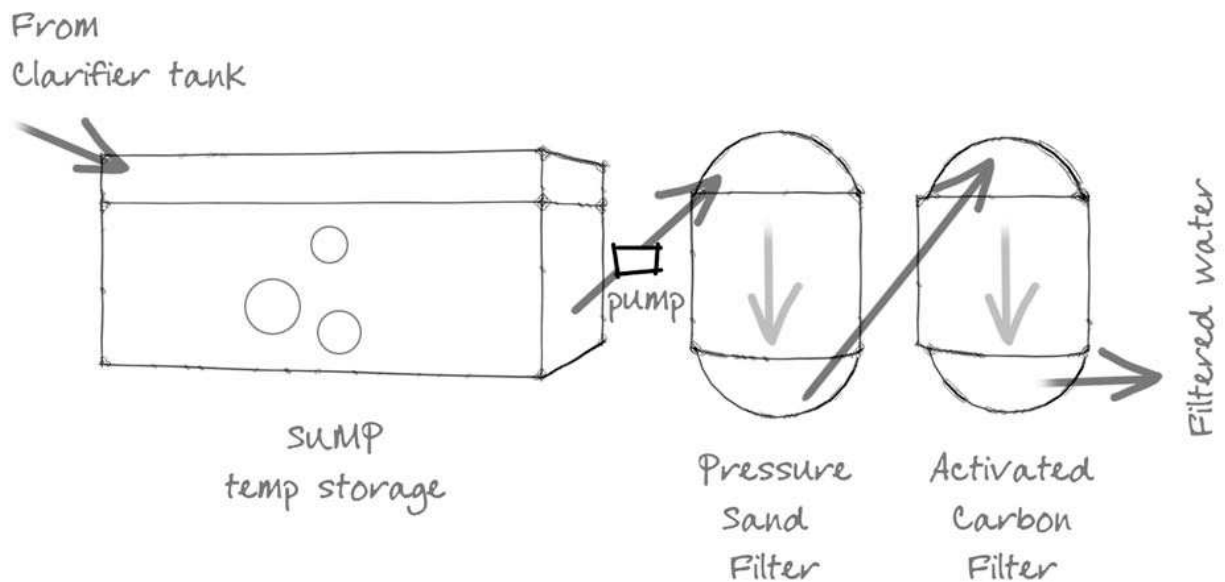
The option is left entirely to the designer/ engineer, provided the rest of the STP has been designed and engineered to the satisfaction of a purist.

9.4 Operation And Maintenance Considerations

- Switch between the main and standby pump every 4 hours (approximately).
- Check oil in the pump every day; top up if necessary
- Check motor-to-pump alignment after every dismantling operation
- Check condition of coupling and replace damaged parts immediately
- Check for vibrations and tighten the anchor bolts and other fasteners
- Check condition of bearings, oil seals, mechanical seal and replace if necessary
- Completely drain out oil and replace afresh as per manufacturer's recommendation
- Always keep safety guard in its proper position
- Follow the LOTO safety principles²⁰ while performing maintenance activities
- Ensure discharge of raw sewage into the aeration tank is visible and can be monitored
- Maintain the flow rate at designed level (no tampering with the bypass valve)
- Follow the manufacturer's O&M manual diligently.

9.5 Troubleshooting

Problem	Cause
Excessive noise	Poor engineering/maintenance
Excessive vibration	Poor engineering/maintenance
Overheating	Poor maintenance
Loss in efficiency of pumping	Poor maintenance



Pressure Sand Filter (PSF)

10.1 Function

The pressure sand filter (PSF) is used as a tertiary treatment unit to trap the trace amounts of solids which escape the clarifier, and can typically handle up to 50 mg/l of solids in an economical manner. This unit is essentially a pressure vessel that is filled with graded media (sand and gravel).

The water filtered with PSF is passed on to the next stage in the STP chain: the Activated Carbon Filter. 21 Stated differently, the pressure-drop across the filter rises sharply



10.2 How It Works

The upper layers of the sand perform the actual filtration function. The gravel layers merely provide physical support to the upper sand layers. The sand used in the PSF is not ordinary construction sand: It has particle size in a specific range, and is specially sieved for this purpose. Think of a sand filter as a 3D ("in depth") filter, as compared to planar filters like a tea bag or tea strainer. Here, the filtration occurs along the entire depth of the sand layer. The solid particles in the water get entrapped and enmeshed in the spaces between the sand particles. Gradually, the space between sand particle gets filled with incoming solids. This blocks the passage of water through the sand layer. As a result, the pressure at the outlet drops rapidly²¹, and wastes the pumping power, and reduces the throughput of the filter. When the pressure drops beyond a limit, the sand is cleaned by backwashing of the filter (back flushing) with water, in which water is passed in the reverse direction (from outlet to inlet). This process agitates, fluidizes and expands the sand bed. The backwash water carries away the lighter pollutant solid particles as backwash waste.

10.3 Design Criteria

A good average design filtration rate is 12 m³/ m²/hr of filter cross-sectional area, and most filters used in STP applications are designed on this basis.

10.4 Construction And Engineering

The Filter vessel is designed as a pressure vessel (it consists of a straight cylindrical shell, with convex dish-shaped ends welded to the top and bottom). A typical vessel is designed to withstand a pressure of 5 kg/cm². In small diameter vessels, it is customary to provide a bolted dish at the top for ease of maintenance. In large filters, a manhole of > 0.6 m dia is provided at the top. A hand-hole of > 200 mm dia is provided at the bottom of the cylinder, to facilitate removal of media from the vessel at the time of servicing. A set of pipes, valves, bypass line, backwash waste line etc. are also provided to facilitate operations such as filtration, bypass (during servicing), backwash etc. Pressure gauges are provided at the inlet and outlet, to monitor the pressure drop across the filter. The shell height typically varies between 1.2 to 1.5 m in small plants. Graded pebbles ranging from 0.5" to 1" are filled as bottom layers in the filter, up to a depth of nearly 0.5 -0.6 m. The top layers consist of the filtering sand media (Coarse and fine sand) to a depth of 0.6 – 0.7 m. A freeboard of nearly 0.3 m above the level of sand may be provided (to allow for expansion of sand during backwash). A great majority of filters operate in the down flow mode (water flowing in top-to-bottom direction). Necessary appurtenances are provided at the top for distributing the inflow uniformly across the cross-sectional area of the filter: similarly, a pipe manifold with laterals is fitted at the bottom as the under drain system. (Without these structures, the water flow inside the filter will be restricted to the center line; and the media placed near the wall of the tank will not contribute to the filtering action.) These good engineering practices ensure optimum filtration efficiency by avoiding short circuiting of flow inside the filter, and also minimizing pressure loss in the filter due to sudden expansion/ constrictions in the fittings.

10.5 Operation And Maintenance Considerations

The operations essentially consist of a long filtration run, followed by a short backwash sequence. The filter needs backwash when the pressure drop across the filter exceeds 0.5 kg/cm². However, it is a good practice to backwash once in a shift, irrespective of the actual amount of pressure loss. A five to ten minute backwash will typically rid the filter of all accumulated muck.

10.6 Troubleshooting

Problem	Cause
Excessive pressure drop across filter	Poor operation/ maintenance
Progressively poor filtration efficiency	Poor operation/maintenance
Very low pressure drop across filter	"Mud balls" formed in filter, discrete sand particles have agglomerated, leading to poor filtration

Activated Carbon Filter (ACF)

11.1 Function

An activated carbon filter, like the Pressure Sand Filter, is a tertiary treatment unit. It receives the water that is already filtered by the Pressure Sand Filter and improves multiple quality parameters of the water: BOD, COD, clarity (turbidity), color and odor.



11.2 How It Works

This filter uses the adsorption action of activated carbon. Activated carbon is typically manufactured from coconut shell or charcoal, the “activation” process creating a highly porous material with a very large surface area. Organic pollutant molecules are physically adsorbed and held fast within the catacomb-like porous structure of the activated carbon. Granular activated carbon is typically used for this purpose. The water filtered by the Pressure Sand Filter enters the Activated Carbon Filter. Unlike in the case of the sand filter, trapped molecules in the carbon cannot be backwashed and got rid of. Hence, activated carbon in the filter has a finite capacity to absorb and hold the pollutants, after which the carbon is said to be exhausted. The exhausted material is removed from the filter and disposed off: Fresh activated carbon is charged in the filter.

11.3 Design Criteria

Very precise design criteria are available for design of activated carbon columns (adsorption isotherms, kinetics of mass transfer between the liquid and solid phase, breakthrough curves etc.). For everyday applications, however, the simplified rules used for the sand filter have been found to be adequate. However, we recommend that the diameter of the activated carbon filter be selected to be 25% larger than the sand filter (SPF) to reduce the frequency of servicing.

11.4 Construction And Engineering

Construction and engineering of the Activated Carbon Filter is similar to the PSF. In addition, on the inside of the filter, epoxy paint coating is recommended due to both the abrasive and corrosive nature of Activated Carbon.

11.5 Operation And Maintenance Considerations

Just as the PSF, the ACF also needs to be backwashed, albeit at a lesser frequency to dislodge any solid particles trapped by simple filtration action. When the carbon gets exhausted (indicated by no improvement in water quality across the ACF), fresh carbon needs to be filled into the filter.

11.6 Troubleshooting

Problem	Cause
Excessive pressure drop across filter	Poor operation/ Maintenance
Treated water smells, or has a color	Carbon life exhausted (change)
Poor BOD/COD	Carbon life exhausted (change)
Black carbon particles in outlet	Poor quality of granular Carbon

Disinfection Of Treated Water

12.1 Function

The treated water is disinfected to destroy and render harmless disease-causing organisms, such as bacteria, viruses, etc. The most common methods of disinfection include Chlorination, Ozonation and UV radiation. Of these, Chlorine finds widespread application. The primary action of the chemical involves damaging the cell wall, resulting in cell lysis and death. In most STPs, the common form of Chlorine used is Sodium Hypochlorite (Hypo) available commercially at 10-12 % strength, being safe, easy to handle and having a reasonable shelf life.

12.2 Design Criteria

Efficiency of disinfection is dependent both on the residual concentration of the chemical used, as well as the contact time, a factor measured as $R \times T$. Generally, a contact time of 20-30 minutes is recommended to achieve over 99 % germicidal efficiencies.

12.3 Construction And Engineering

The Chlorine disinfection system consists of a Hypo-holding tank (its size depends on the flow rate of the STP) and an electronically metered dosing pump. Hypo solution of desired concentration is prepared in the tank. The dosing rate is set in the metering pump as per the desired Chlorine dose rate, typically 3-5 PPM. Hypo solution is dosed at the outlet of the ACF, online, so that adequate mixing of Hypo with the treated water is achieved.

12.4 Operation And Maintenance Considerations

Prepare fresh Hypo solutions every day in the day tank Shelf life of over 2 months is not recommended, especially during summer Store Hypo in a cool place Study Material Safety Data Sheets of Hypo and follow instructions Periodically check available Chlorine strength of Hypo Check and record Residual Chlorine concentration every day

12.5 Troubleshooting

Problem	Cause
Inadequate Chlorine residual in treated water	Poor operation/ poor quality chemical
Fishy smell in treated water	Excessive chlorination

Excess Sludge Handling

13.1 Function

Biological treatment of wastewater performs produces excess biological solids due to the growth and multiplication of bacteria and other microorganisms in the system. The excess biomass thus produced needs to be bled out of the system, and disposed off efficiently. This is a five-step process: sludge removal, storage, conditioning, dewatering and disposal. Sludge is removed ("bled") from the system from the sludge recirculation pipeline (through a branch). The sludge is in the form of a thick slurry. It is taken into a sludge-holding tank, and kept under aeration (to prevent the living organisms from putrefying) until dewatering operations can be carried out. Before dewatering, polymer or other chemicals may be added for conditioning the sludge, to facilitate the process. Sludge is then de watered din a filter press/ Sludge bag/ centrifuge.

13.2 Design Criteria

The quantity of excess sludge generated in the STP is dependent on various factors including the BOD concentration, MLSS levels, temperature etc. The F/M loading rate is however is the factor which chiefly determines the amount of excess solids produced. Sufficient data are available in literature in graphical form for determination of this number. A typical figure for use in India is between 0.20 to 0.25 times (on dry mass basis), kg of BOD removed in the aeration tank, in extended aeration systems with low F/M. Since the excess sludge is available in slurry form from the sludge recirculation line, the slurry consistency may be taken to be between 0.8 to 1.0 %. Simple arithmetic using the above two numbers gives the quantity of excess sludge to be handled per day in m³/day.

13.3 Construction And Engineering

We will discuss two different methods of handling the dewatering system:

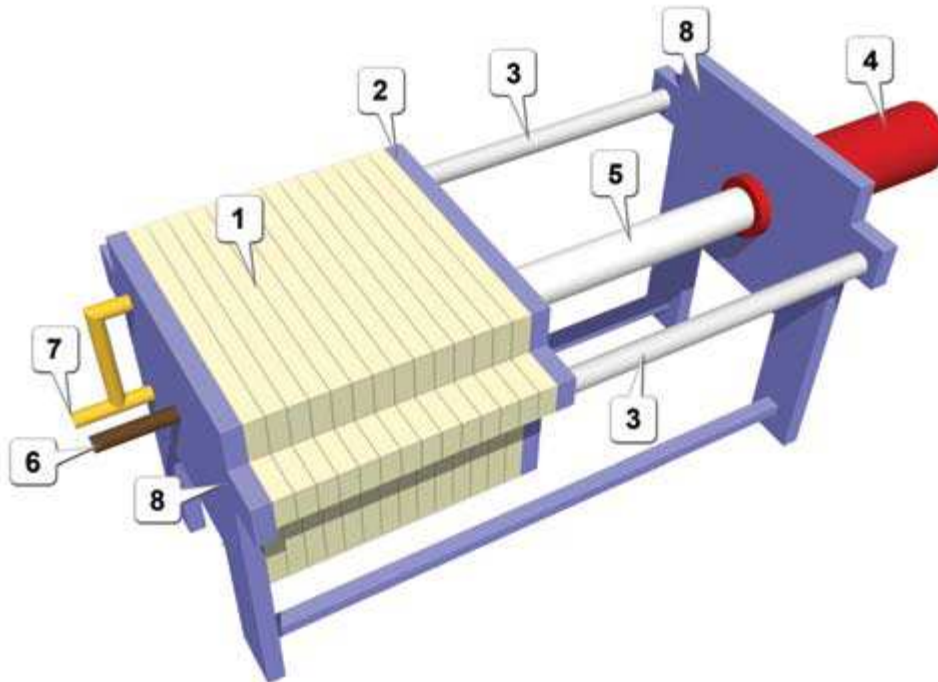
- Plate-and-Frame Filter press
- Bag-type dewatering

13.3.1 Plate-and-Frame Filter press

The Plate-and-Frame Filter press is the most commonly used method of dewatering the sludge. It consists of three to four parts:

- Sludge-holding tank with aeration/ mixing
- Polymer solution-preparation tank and dosing (for conditioning the sludge)
- High-pressure filter-press feed pump
- Plate-and-frame filter press

The polymer solution tank must be of sufficient capacity to hold 0.1 % solution of polymer to condition one batch of excess sludge: Typically, polymer requirement is between 1-2 % of the excess sludge on dry weight basis. The sludge-holding tank must be of sufficient capacity to accommodate the combined volume of (a) the excess sludge to be dewatered in a single batch, and (b) the polymer solution that is added. The high-pressure filter press-feed pump (4 – 5 kg/cm² pressure) must be selected to dewater a single batch of excess sludge within 3-4 hours of filter press operation. A typical press is shown on the next page.



SL	Remarks
1	The press makes use of multiple filter plates. Each plate has indentations and recesses on the sides, such that when two plates are pressed together, a hollow chamber is formed between the plates. The dewatered sludge settles in this hollow part to form a brick (also called “cake”).
2.	The end-plate is a solid plate used to press the filter plates (1) together.
3.	All the filter plates and the end-plate have “wings”, which rest on these two rails. All the plates hang on the rails, and they are designed to easily slide along the rail when pushed.
4.	The figure shows a hydraulic jack that extends and retracts the plunger
5.	The job of the plunger is to tightly press the filter plates (1) against each other, so that the slurry (which is injected in the press under high pressure) does not leak from the joints between the filter plates. In smaller filter presses, the electric jack and plunger are replaced with a large screw that is turned manually with a large wheel.
6.	The inlet pipe brings the excess sludge slurry from the screw pump (not shown in the figure). The slurry travels the path of least resistance and fills up the cavities between the filter plates (1). All cavities are lined with a filter cloth
7.	The liquid in the slurry (called “filtrate”) can pass through the cloth, while the thicker paste-like sludge (mainly bacterial flocs and some solids) cannot pass through the cloth. The filtrate travels to the outlet ports at the top and bottom of the plate, and from there it is discharged through the filtrate discharge pipe. The solids remain in the cavity between the plates. Typical thickness of cake in the chamber is 2 to 5 cm, A freshly produced cake has a moisture content of between 70 – 75 %. Note that it is impossible to achieve a bone-dry cake even after prolonged air-drying: The moisture content would not drop below 50%. In any case, the main use of the cake is used for making compost, in which case the sludge has to be moistened once again. Therefore the cake is usually not dried further. Once the filtrate stops coming out of the discharge pipe (7), the jack (4) is deactivated, which retracts the plunger (5). Now the filter plates (1) are slid apart on the rails (3). As soon as the plates separate, the cakes fall down in a basket placed below the press. Now the press is ready for another cycle.
8.	The work-bench is simply a support structure for the parts described above

13.3.2 Bag-type dewatering

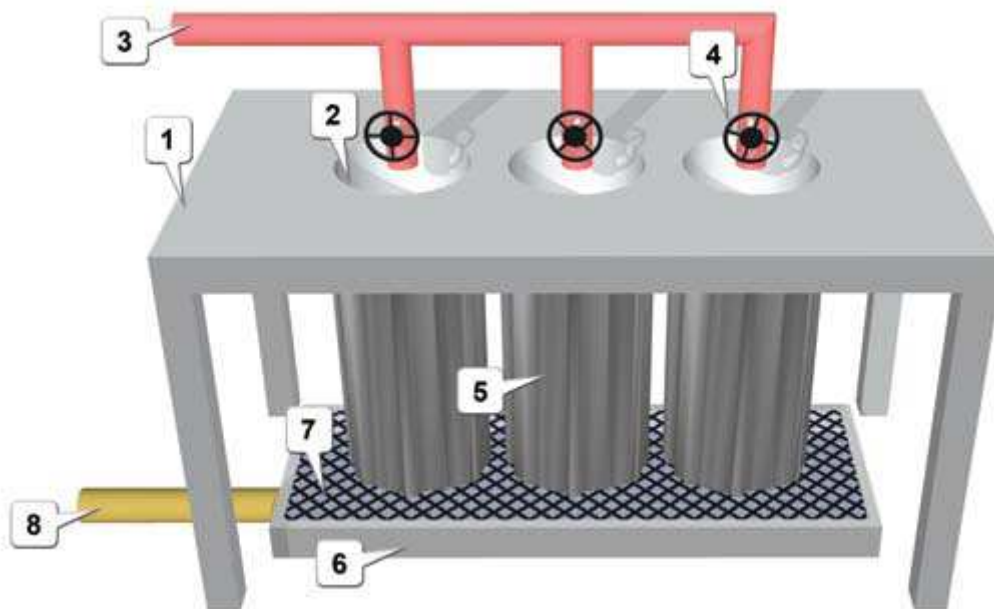
This contraption actually is an imitation of our daily tea-strainer! In fact, once the sludge is strained, it does look remarkably like used tea powder (not the CTC or whole tea leaves, but the lowly tea dust.)

The key factor that allows this method is that the flocs of bacteria are globular with 2-3 mm diameter. These flocs can be easily trapped with a laminated poly bag. These bags are used to transport sugar, fertilizer and other foodstuff, and at the end of the cycle, they are discarded. These used bags are available in secondary market at very cheap rates. The equipment is very simple, as shown on facing page: A rustproof Stainless Steel table is manufactured with provision to hang 2-4 bags. The sludge is poured in each bag, and just like tea-strainer, the water drains out.

SL Remarks

- 1 The stainless steel table allows splashing of water without getting rusted.
- 2 The openings have a collar beneath, where the bags can be clamped
- 3 The excess sludge pipeline brings sludge from the settling tank.
- 4 The valves can be used to regulate the flow rate, and also to turn off if there is no bag beneath.
- 5 The laminated poly bags capture the floc, but allow the water to run.
- 6 The stainless steel collection tray collects the water (and fine sediments)
- 7 The stainless steel grill supports the bags and allows the water to fall through
- 8 The collection pipe takes the filtrate back to the Equalization tank. (Sump and pump required for this purpose are not shown).

However, because the sludge is not squeezed (as in the frame-and-plate press), the bag remains wet. So, after the water stops dripping, the operator has to close the bag and place it on airing racks to let it dry for several days. In fact, if the dewatered sludge is to be used as compost, then it need not be dewatered further: The moist sludge can be directly used as an ingredient!

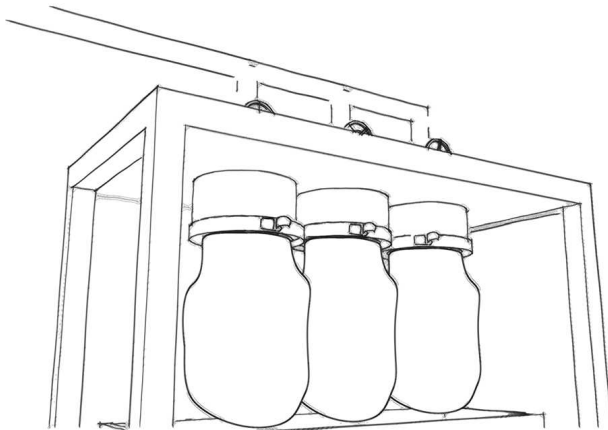


13.4 Operation And Maintenance Considerations

Typically, in small plants, the filter press may be sized for a single batch operation per day. In large plants, three batches per day, one per shift, is the norm. Fresh sludge (not more than a day old), kept fully aerated and mixed (agitated), dewateres easily in the filter press. Hence, sludge must not be stored in the holding tank for longer durations. The desired quantity of polymer needs to be added 15 - 30 minutes before the dewatering operation. Filter press operation is carried out over 3 - 4 hours, or when filtration ceases. After every dewatering operation, the filter cloths must be thoroughly cleaned, so that clogging in the pores of the woven polypropylene filter fabric is avoided. Periodic cleaning of filter cloth with Hypo solution will also prolong the life of cloth. When the filtration process becomes excessively slow, it is time to replace the filter cloth with a fresh set. Normal maintenance as prescribed by the manufacturer may be practiced for the high pressure helical screw pump. Care must be taken not to damage the rubber stator of the screw pump by dry running of the pump. It is generally preferable to locate the pumps such that positive suction is enabled.

13.5 Troubleshooting

Problem	Cause
Filter press does not dewater the slurry sufficiently quickly	Poor design/ engineering
Dewatering is very slow	• Oily / slimy sludge, • Filter cloth is clogged, • Improper conditioning • Incorrect filter cloth fabric
Insufficient pressure developed	• Rubber stator of screw pump worn out • Steel rotor damaged



Miscellaneous Considerations

- Genset backup power to run the entire STP in case the mains electricity line fails
- Design and engineer the STP to be operator friendly
- Design and engineer the STP for the operator's safety, health and hygiene
- Adequate illumination in STP if in a room, or basement
- Totally covered, underground STPs are neither operator-friendly, nor maintenance-friendly
- Adequate exhaust and ventilation system to be provided for operator comfort, health and hygiene
- Without proper exhaust/ ventilation in enclosed spaces, Carbon Dioxide accumulates, gets converted to Carbonic acid and corrodes metallic parts in the STP. Carbonic acid also depresses pH of the wastewater, thus affecting treatment performance
- Provide safe and comfortable access to all units in STP for monitoring, operation, and maintenance
- Prepare and maintain a Standard Operating Procedure for the STP and train all operators to follow those procedures
- Prepare and maintain an operating log book for all activities in the STP
- Prepare and maintain a mechanical checklist for routine preventive maintenance
- Prepare and maintain a History Sheet for each critical equipment in the STP
- Prepare and maintain a chemicals/ consumables stock register
- Periodically check and validate all log books, checklists etc.
- Provide a Water meter at the outlet of the ACF to monitor average daily throughput of the STP
- Ventilation system: Generally 10-12 air changes per hour should give good ventilation²². The air change is calculated based on the open/ vacant head space of the underground/ basement room. There must be a fresh air fan (forced draft) and an exhaust fan (induced draft), with two separate ducting systems.
 - If the induced draft fan is designed for a slightly higher capacity than the fresh air fan, then the room will always be under a slight negative pressure, and gases will not escape the room as fugitive emissions.